

Graphics Agent for Linux

Administrators' Guide

26.05



Last updated: Thursday, July 16, 2026

Overview

Anyware Graphics Agent for Linux 26.05

The Graphics Agent for Linux is part of HP Anyware. It enables users to deliver GPU-powered physical and virtual workstations to end users via remote clients.

Users include high-end knowledge workers, graphic designers, artists, and CAD/CAM designers.

A deployed Graphics Agent for Linux requires these components:

- **A host machine** which provides the desktop to remote clients. The host must be a virtual machine in a data center or in the cloud. See [System Requirements](#) for more information.
- **The agent software** installed on the host machine.
- **A GPU.** The Anyware Graphics Agent requires an NVIDIA GRID card.

The GPU can be dedicated to the workstation, or shared among virtual workstations by a hypervisor. See [System Requirements](#) for more information.

Where to Find Information about Other Components

This guide describes the Graphics Agent for Linux.

For more information about Anyware clients, see one of the following:

- [Software Client for Windows Administrators' Guide](#)
- [Software Client for macOS Administrators' Guide](#)
- [Software Client for Linux Administrators' Guide](#)
- [Zero Client Firmware Administrators' Guide](#)

For information about Cloud Access licensing, see our [Licensing FAQ](#). Most PCoIP systems use PCoIP Cloud Licensing. For systems using a local PCoIP License server instead, refer to the following guides:

- [PCoIP Linux License Server Administrators' Guide](#)
- [PCoIP Windows License Server Administrators' Guide](#)

For information about HP Anyware licensing, see our [Licensing FAQ](#). Most Anyware systems use Cloud Licensing. For systems using a local License server instead, refer to the following guides:

- [PCoIP License Server Administrators' Guide](#)
- [PCoIP Windows License Server Administrators' Guide](#)

What's New in This Release

Release 26.05 of the Graphics Agent for Linux contains bug fixes, security fixes, and stability enhancements. There are no new features in this release.

System Requirements

The Graphics Agent for Linux depends on the following system capacities and capabilities:

Compatible Host Instance Types

Environment	Supported instance types
esxi ESXi (6.0+)	VMware Hardware Version 11
KVM (5.0+)	QEMU/KVM
Physical machines	Linux physical machine ¹
AWS EC2	• G4 • G5
Microsoft Azure	• NV-series • NVv3-Series • NCasT4_v3-Series

Info

Starting with 25.06, HP Anyware does not test on Google Cloud Platform. However, it is expected to remain fully functional.

Host Instance Requirements

Info

- Support for CentOS 7 and RHEL 7 was removed in HP Anyware 2024.03.
- Active support for RHEL 8 and Rocky 8 ended in May, 2024. HP Anyware version 25.06 was the last release supporting these operating systems.
- As a general policy, HP Anyware will support RHEL and Rocky versions during their active support phase.
- Ubuntu releases will only be supported until standard support ends from Ubuntu.

Global instance requirements	
Operating Systems	• RHEL/Rocky Linux 9 • Ubuntu 22.04 LTS, 24.04 LTS • Alma Linux 9 HP Anyware has not been tested on Alma, but is expected to be fully functional. For troubleshooting, issues should be reproducible on RHEL or Rocky Linux and not only limited to Alma OS.
Remote Host Memory	At least <i>2GB</i> of RAM is required on the host desktop. The agent should have at least <i>512MB</i> of available memory.
Remote Host GPU Memory	At least 1GB <i>per 4K UHD display</i>
Remote Host CPUs	At least 2 CPUs are required on the host desktop. Processors must support Streaming SIMD Extensions (SSE) 4.2. <i>To use PCoIP Ultra, processors must support the AVX2 instruction set.</i>
Network Ports	The following ports must be open on the host desktop: • TCP 443 • TCP 4172 • UDP 4172 • TCP 60443 Collaboration sessions require 5 open UDP ports (64172-64176) UDP port 4172 does not need to be opened when using Anyware Manager Enterprise.
Storage	At least 250MB for installation and 100MB for logging are recommended.
User	Cannot be <code>root</code> . You must create a user account for PCoIP connections.

 Note: Rocky Linux OS support lifecycle

Rocky Linux versions are only supported until active support ends from Rocky Linux.

Supported GPUs

Instance	GPU	Driver	Driver download instructions
VMware ESXi	GPUs as listed in NVIDIA's GRID release notes .	GRID 17.1+	Get drivers
KVM	GPUs as listed in NVIDIA's GRID release notes .	GRID 17.1+	Get drivers RHEL only
Physical PC <i>NVIDIA GPUs</i>	• Tesla (any) • Quadro (k2000+)	Use the latest 580.126.09+ driver for your supported GPU	Get drivers
AWS EC2 G4dn	NVIDIA T4	GRID 19.4+	
Get drivers			
AWS EC2 G5	NVIDIA A10G	GRID 19.4+	
Get drivers			
Microsoft Azure	NVIDIA M60, T4	GRID 18.6+	Get drivers

Info: Display limitations on Microsoft Azure

Graphics Agents running on Microsoft Azure may be limited to a single 2560x1600 display, depending on the state of NVIDIA GRID licensing.

Note: NVIDIA Tesla cards have license limitations

Systems using NVIDIA Tesla-based GPUs, including M60, M10, and M6, are subject to licensing restrictions which may further reduce the available number of monitors or their available resolutions. For systems using Tesla-based GPUs, only licensed displays and resolutions will use GRID functionality. For more information, see [NVIDIA's Grid Licensing User Guide](#)

Certificate Requirements

Your OS's Trusted Root Certificate Authorities must include **Amazon Root CA 1-4** for Cloud License Service (CLS) to function properly. Licensing issues may occur if they are missing.

For detailed instructions on installing Amazon Root CA 1-4, see [Solution to Licensing Problems with Cloud License Service \(CLS\)](#).

- **1** Installation on physical machines requires some additional configuration. See the extra installation instructions for Ubuntu and RHEL/CentOS.

Feature Support

Audio

Audio Support

The Graphics Agent for Linux supports audio input and output between the host and the client. Audio can be enabled or disabled and audio bandwidth can be throttled by [configuring the agent](#).

Note: Automatic Selection

All available audio input and output devices should be automatically selected and enabled by default when the client initially connects to a session with the HP Anyware Audio Driver.

Collaboration

PCoIP Ultra Collaboration enables a PCoIP session user to share their session with multiple remote collaborators using Anyware Software Clients.

Note: Collaboration terminology

When discussing this feature, we will refer to the first user as the *session owner*, and subsequent users who join the session as *collaborators*. The session owner's screens, audio, and input devices (if allowed) are shared with the collaborators when they join the session.

Up to 5 collaborators can join an ongoing PCoIP session using the same invitation. The maximum number of collaborators can be reduced by [configuring the agent](#).

Warning: Consider system resources

As the number of collaborators increases, the load on the Anyware agent's CPU, memory, and other resources will also increase. Test your system to ensure it can support all of your planned collaborators.

While connected, all collaborators can view and hear the session owner's screens and audio, and see the controlling collaborator's mouse movements. If permitted by the session owner, they can also share control of the session owner's keyboard and mouse using [input control](#).

During a collaboration session, *all* of the session owner's desktop screens may be shared depending on the session owner's Anyware Software client display settings. See [Understanding Display Behavior](#) for more information.

Collaboration Requirements

PCoIP Ultra Collaboration is supported by all Anyware agents. Anyware Software clients that support PCoIP Ultra can participate in collaboration sessions (all Anyware Software clients 23.04 and higher meet this requirement).

Some collaboration features have specific version requirements for Anyware agents and Anyware Software clients; [these are noted below](#).

Note: Only Anyware Software Clients are Supported

Anyware Tera2 Zero Clients and mobile clients do not support PCoIP Ultra and cannot join collaboration sessions.

Feature Version Requirements

PCoIP Ultra Collaboration features depend on coordinated updates in Anyware agents and Anyware clients, so review these requirements carefully to ensure the features you need are supported by your system. **Use latest versions of both Anyware agent and Anyware client.**

Feature	Required versions	Notes
Collaborate menu	Anyware agent 23.06+ Anyware client 23.06+	Both client and agent must be 23.06 or higher, with Collaboration and PCoIP Ultra enabled.
Multiple collaborators	Anyware agent 23.04+ Anyware client 23.04+	HP Anyware versions 22.07–23.01 supported single collaborators only.
Input control	Anyware agent 23.01+ Anyware client 23.01+	See Input Control for more information.

Network Requirements

Each collaborator connection requires a separate UDP port. These ports are assigned in a range that *begins* with the configured UDP port (by default, 64172), and increments with each additional collaborator. All ports in this range must be open, both at the cloud provider network level and the local firewall at the host.

For example, using the default configuration and hosting three collaborators, the system would require inbound UDP ports 64172, 64173, and 64174 to be open.

Maximum number of collaborators	Required inbound UDP ports
1	64172
2	64172-64173
3	64172-64174
4	64172-64175
5 (default)	64172-64176

You can [change the starting port number](#) if desired. If you change the configured starting UDP port, adjust these ranges and ensure your host firewall configuration allows traffic on the new ports.

For direct connections, or brokered connections that do not use a Security Gateway, each collaborator's Anyware client must be able to reach these ports. For brokered connections using a Security Gateway, only the Security Gateway must be able to reach them.

Enabling Collaboration

⚠ Important: Anyware agent steps

Collaboration sessions are enabled and configured on the Anyware agent machine before starting collaboration sessions. Make sure the Anyware agent version you are using supports the collaboration features you expect. For details, see [Feature Version Requirements](#).

PCoIP Ultra Collaboration is enabled, disabled, and configured on the Anyware agent machine. It is disabled by default, and must be enabled by activating both *PCoIP Ultra* and *Collaboration* on the remote desktop.

The following steps apply to the session owner's desktop machine.

To enable PCoIP Ultra Collaboration:

1. Open `/etc/pcoip-agent/pcoip-agent.conf` in a text editor.
2. If PCoIP Ultra is not already enabled, enable it by adding a new line in the text editor, and configuring the PCoIP Ultra offload mode:

PCoIP Ultra mode	Supported by	Add a line with this value
CPU Offload	Standard Agents, Graphics Agents	<code>pcoip.ultra = 1</code>
GPU Offload	Graphics Agents	<code>pcoip.ultra = 2</code>
Auto Offload	Graphics Agents	<code>pcoip.ultra = 3</code>

3. Add a new line enabling collaboration:

```
pcoip.enable_collaboration = 1
```

4. **Optional:** To enable *Collaboration Input Control* if desired, stay in the text editor and follow the instructions in [Enabling Input Control](#) below.
5. **Optional:** You can change the UDP starting port if needed (the default starting port is UDP 64172). To change the collaboration port number, add a new line specifying the new value:

```
pcoip.collaboration_udpport = <new_collaborator_port>
```

...where `<new_collaborator_port>` is your new starting port number.

6. Save the file and exit the editor.
7. Restart the Anyware agent service:

```
sudo systemctl restart pcoip
```

See [Configuration Guide - Configurable Settings](#) for more detailed information on setting configuration values.

Input Control

Collaborator input control allows collaborators to use their own mice and keyboards to control the session owner's desktop. **This feature is disabled by default**, and must be enabled on the Anyware agent before it is available.

Enabling Input Control

Input control is disabled by default, and the option to give collaborators input control cannot be selected.

To use Input Control, enable it on the Anyware agent. This change takes effect on the next PCoIP session:

1. Using a text editor, open the following file:

```
/etc/pcoip-agent/pcoip-agent.conf
```

2. Add a new line enabling input control:

```
pcoip.enable_collaboration_input_control = 1
```

3. **Optional:** Provide a custom input control timeout value (specified in milliseconds; 3000ms is 3 seconds):

```
pcoip.collaboration_input_control_timeout <timeout_value_in_ms>
```

4. Save the file and exit the text editor.

Disabling Input Control

If Input Control has been enabled and you wish to disable it again:

1. Using a text editor, open the following file:

```
/etc/pcoip-agent/pcoip-agent.conf
```

2. Disable input control:

```
pcoip.enable_collaboration_input_control = 0
```

3. Save the file and exit the text editor.

Configuring Collaboration

Changing the Collaboration Starting Port

The default *starting* UDP Port for collaborator sessions is 64172. You can change this value if needed. Remember that you must also open a range of UDP ports that *begin* with this value to accommodate all of your collaborators; see [Network Requirements](#) for examples.

To change the Collaboration session port:

1. Open `/etc/pcoip-agent/pcoip-agent.conf` in a text editor.
2. Create new config entry specifying the new starting UDP port number to use:

```
pcoip.collaboration_udpport = <new_collaborator_port>
```

3. Save the file and exit the text editor.
4. Restart the Anyware agent service:

```
sudo systemctl restart pcoip
```

Changing the Maximum Number of Collaborators

PCoIP Ultra Collaboration supports up to 5 collaborators on the same PCoIP session. You can further limit the number of allowed collaborators by changing the *maximum collaborators* setting to a value from 1-5. By default, the system allows 5 collaborators.

To Change the Maximum Number of Collaborators:

1. Open `/etc/pcoip-agent/pcoip-agent.conf` in a text editor.
2. Create a new entry, replacing `<maximum number of collaborators>` with your desired value from 1-5:

```
pcoip.max_collaborators = <maximum number of collaborators>
```

3. Save the file and exit the editor.
4. Restart the Anyware agent service:

```
sudo systemctl restart pcoip
```

Changing the Input Control Timeout Value

Input control is released and made available to other collaborators by idling all input devices for a brief period. By default, this control timeout is 3 seconds (3000ms). You can change this value by configuring the Anyware agent machine as follows:

1. Using a text editor, open the following file:

```
/etc/pcoip-agent/pcoip-agent.conf
```

2. Provide a custom input control timeout value (specified in milliseconds; 3000ms is 3 seconds):

```
pcoip.collaboration_input_control_timeout = <timeout_value_in_ms>
```

3. Save the file and exit the text editor.
4. Restart the Anyware agent service:

```
sudo systemctl restart pcoip
```

Sharing Your Session With Collaborators

You can invite up to 5 collaborators to participate in your session, and optionally allow them to control your desktop.

⚠ Important: Anyware Client steps

Collaboration sessions are shared from Anyware clients in established PCoIP sessions. Make sure the software client version you are using supports the collaboration features you expect. For details, see [Feature Version Requirements](#).

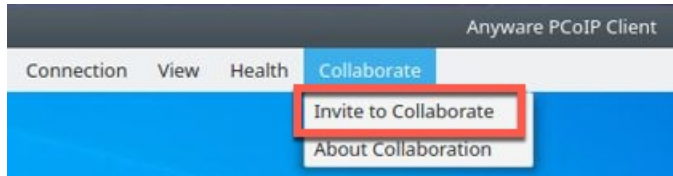
Collaboration sessions are managed using the **Collaboration manager**. The collaboration manager shows you who is connected to your session, whether each collaborators can view or control the session, and allows you to invite new collaborators or stop collaborating.

✎ Note: New Collaboration Manager menu option

The Collaboration manager can now be launched by using the client's in-session menu, in addition to the system menu bar.

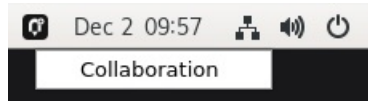
To launch the Collaboration Manager:

1. Connect to a PCoIP session with PCoIP Ultra and Collaboration enabled.
2. From the remote session, open the **Collaboration Manager** using either of these methods:
 - **From the client menu:** Select **Collaborate** > **Invite to Collaborate**.



The Collaborate menu appears if both the Anyware agent and Anyware client are version 23.06 and higher, and if both support Collaboration. If the Collaborate menu does not appear, open the Collaboration Manager using the menu bar instead.

- **From the menu bar:** by clicking the **Anyware Collaboration** icon in the menu bar:



This example shows a Linux desktop; yours may vary depending on which operating system you are connecting to.

To invite collaborators, the session owner generates a *collaboration invitation* using the collaboration manager, and distributes the invitation to all collaborators.

About Collaboration Invitations

Collaboration invitations are created by the session owner and distributed to collaborators, who use them to join an established collaboration session.

Note: About collaboration invitations

A single collaboration invitation can be used by multiple collaborators (up to the maximum number configured). You do not need to generate a new invitation for each collaborator. Collaboration invitations behave as follows:

- If an invitation is generated but no collaborators connect within one hour, it expires and can no longer be used. If this happens, generate a new invitation.
- If *any* collaborators connect using an invitation, the invitation is activated and its time limit is removed. Once activated, an invitation can be re-used until the session owner stops collaborating or ends the session.
- Collaborators can disconnect from a collaboration session and then rejoin it later using the same invitation.
- Collaboration sessions persist even if all collaborators leave and only the session owner remains. Until the session owner disconnects or stops the collaboration session, collaborators can rejoin the session using the same invitation.
- The collaboration invitation remains valid until the session owner disconnects or stops the collaboration session.

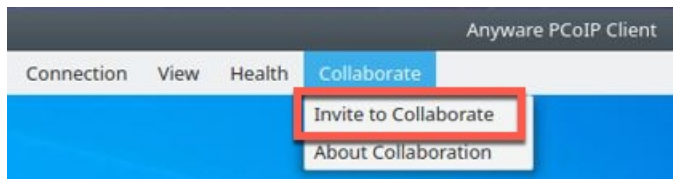
Inviting the First Collaborator

To begin a collaboration session, generate an invitation using the Collaboration Manager.

To generate a collaboration invitation:

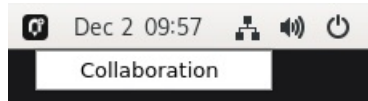
1. Connect to a PCoIP session with PCoIP Ultra and Collaboration enabled.
2. From the remote session, open the **Collaboration Manager** using either of these methods:

- **From the in-session menu:** Select **Collaborate > Invite to Collaborate**.



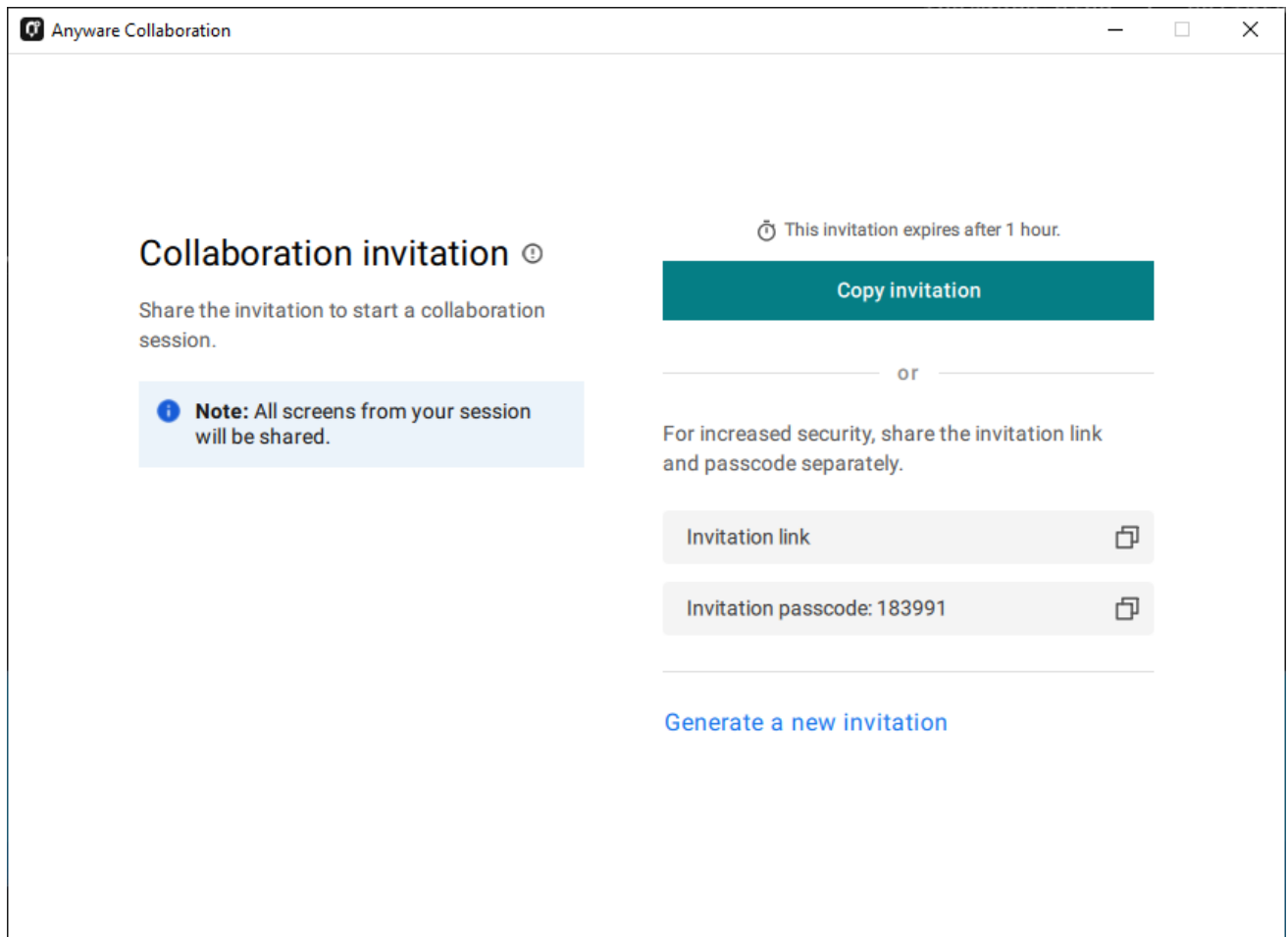
The Collaborate menu appears if both the Anyware agent and Anyware client are version 23.06 and higher, and if both support Collaboration. If the Collaborate menu does not appear, open the Collaboration Manager using the menu bar instead.

- **From the menu bar:** by clicking the **Anyware Collaboration** icon in the menu bar:



This example shows a Linux desktop; yours may vary depending on which operating system you are connecting to.

3. The Collaboration manager generates and displays an invitation:



The invitation contains two pieces of information that are used to invite the collaborator:

- **Invitation Link:** The collaborator will use this link to join your session. The link may be opened on any Mac, Windows or Linux machine with a Anyware Client 21.07 or newer.
- **Invitation passcode:** This is a 6-digit code that confirms the identity of the individual connecting to the collaboration session. A new code is generated along with each new token.

4. Share the *invitation link* and the *invitation passcode* with the collaborator.

- To share both the link and the code at once, click the **Copy invitation** button. This will create a single message containing both the link and the code and place it on your clipboard. Share this with your collaborators using any acceptable method.
- To share the link and code *separately*, click the *copy* button beside each item and share them using separate communications. Sharing the invitation this way reduces risk in the event that a message is inadvertently sent, forwarded, or intercepted by a third party.

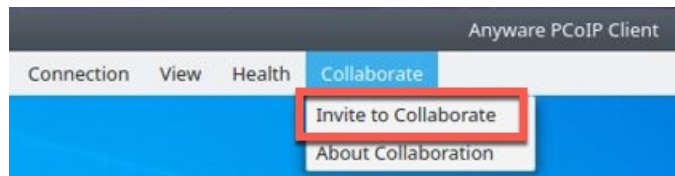
Inviting Additional Collaborators

Once the collaboration session has been created, you can invite additional collaborators by sharing the same invitation link and passcode with them. You can also view the invitation, and copy its link and passcode for sharing, using the Collaboration Manager.

To view and copy the invitation link and passcode:

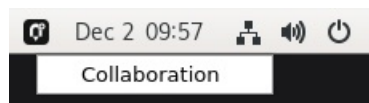
1. From the remote session, open the **Collaboration Manager** using either of these methods:

- **From the in-session menu:** Select **Collaborate > Invite to Collaborate**.



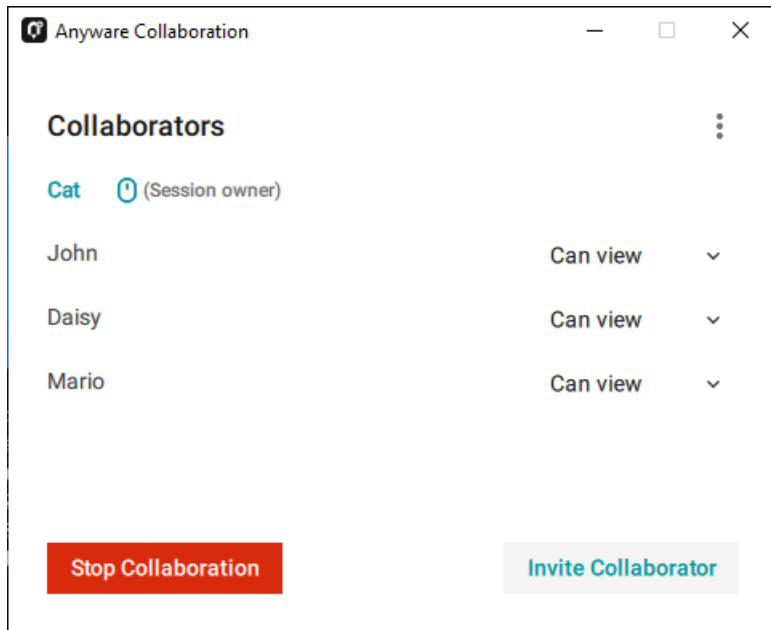
The Collaborate menu appears if both the Anyware agent and Anyware client are version 23.06 and higher, and if both support Collaboration. If the Collaborate menu does not appear, open the Collaboration Manager using the menu bar instead.

- **From the menu bar:** by clicking the **Anyware Collaboration** icon in the menu bar:

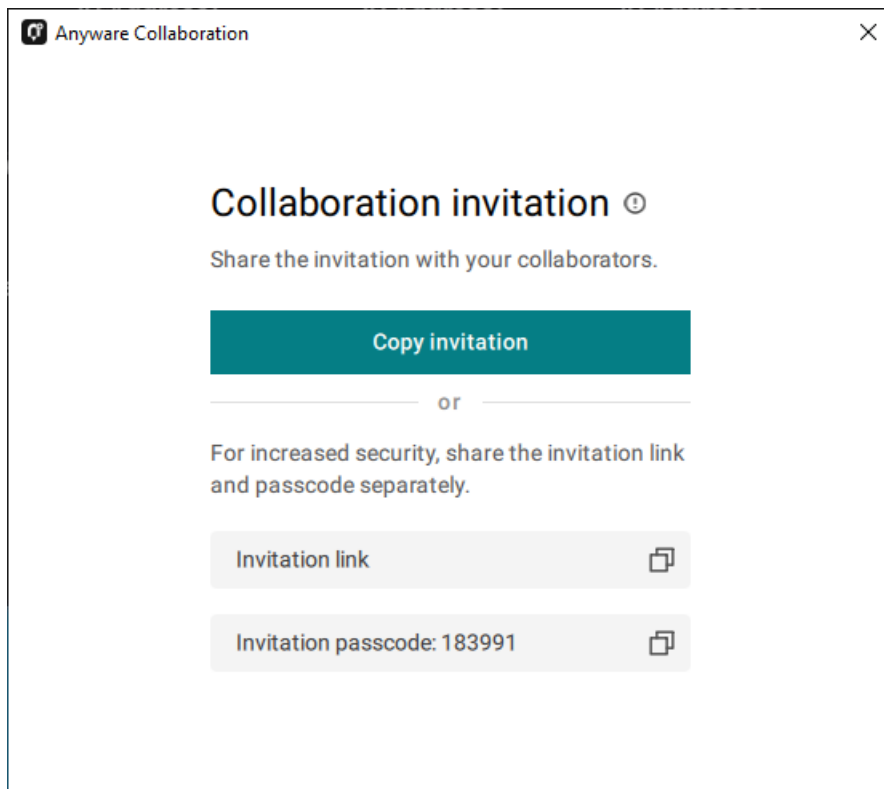


The collaboration manager shows a list of your active collaborators (if any).

2. In the Collaboration manager, below the list of active collaborators, click **Invite Collaborator**:



3. The Collaboration manager displays the generated invitation. Note that this is the *same* invitation link and passcode you used previously. It is not a new invitation:



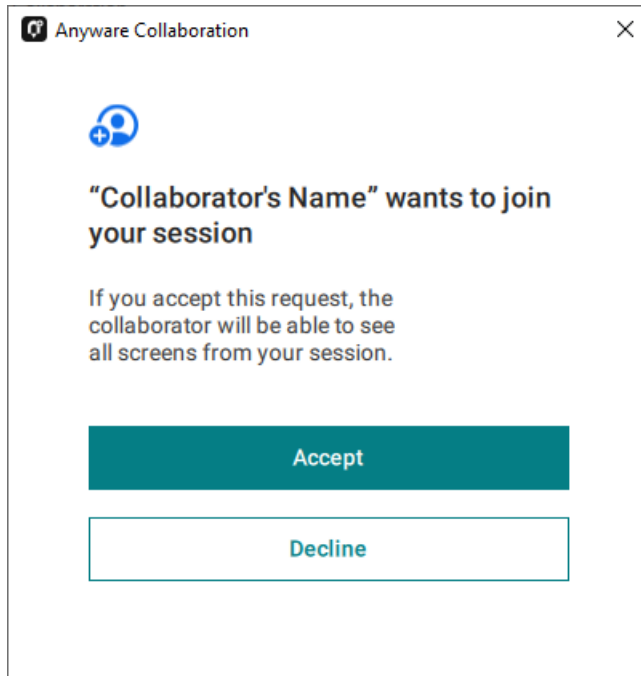
4. Share the *invitation link* and the *invitation passcode* with the additional collaborators.

Accepting or Declining Collaborators

Once distributed, the session owner's collaborators can [join the collaboration session](#). As collaborators use the invitation, the session owner is notified and can accept or reject each connection attempt.

To respond to a collaborator:

1. When the collaborator attempts to join the session, the Collaboration manager will display options to accept or reject the connection.



2. Click **Accept** to start the collaboration session. Click **Decline** to deny the request. Whether you accept the request or not, the invitation has been used and is now disabled. Subsequent attempts will require a new invitation.

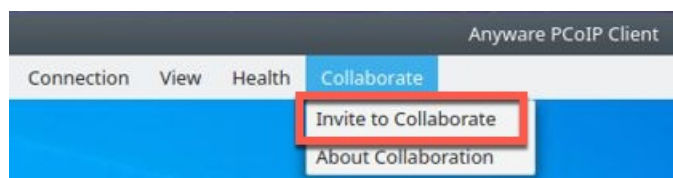
Ending a Collaboration Session

The collaboration session will end when the session owner disconnects their PCoIP session, or if they stop collaborating using the collaboration manager.

Ending the collaboration session invalidates the invitation. To start a new session, generate a new invitation by inviting another collaborator.

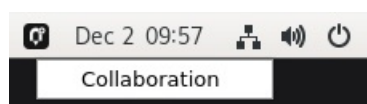
To stop collaborating:

1. From the remote session, open the **Collaboration Manager** using either of these methods:
 - **From the in-session menu:** Select **Collaborate > Invite to Collaborate**.



The Collaborate menu appears if both the Anyware agent and Anyware client are version 23.06 and higher, and if both support Collaboration. If the Collaborate menu does not appear, open the Collaboration Manager using the menu bar instead.

- **From the menu bar:** by clicking the **Anyware Collaboration** icon in the menu bar:



This example shows a Linux desktop; yours may vary depending on which operating system you are connecting to.

2. Click the **Stop Collaboration** button.

Allowing Collaborators to Control the Session

Collaborator input control allows collaborators to use their own mice and keyboards to control the session owner's desktop. **This feature is disabled by default**, and must be enabled on the Anyware agent before it is available.

Once enabled, input control options are available from the collaboration manager. Input control can be granted (or retracted) for each user separately or for all users at once.

Note: Disabling input control globally

You can disable Input Control on the Anyware agent, which turns the feature off entirely. When disabled this way, session owners will not be able to allow collaborators to take control, and all sessions will be view-only. For more information, see [Disabling Input Control](#).

Important: The session owner always has control of their Anyware client's in-session menu

The session owner always has control of their Anyware client's in-session menu. If the owner is unable to reclaim session input control for any reason, they can disconnect the PCoIP session using the in-session menu option. When the owner disconnects from the session, the collaborator is immediately disconnected.

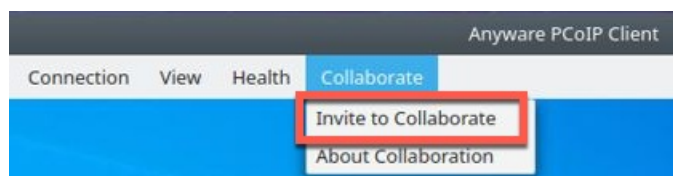
Enabling Input Control for Collaborators

The following steps will allow one or more collaborators to take control of the session desktop. The collaborators will not immediately have control when this is granted; they must still take control using the [following process](#).

This option will not be available if Input Control has been disabled on the Anyware agent.

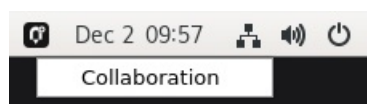
To allow collaborators to control the session desktop:

1. From the remote session, open the **Collaboration Manager** using either of these methods:
 - **From the in-session menu:** Select **Collaborate > Invite to Collaborate**.



The Collaborate menu appears if both the Anyware agent and Anyware client are version 23.06 and higher, and if both support Collaboration. If the Collaborate menu does not appear, open the Collaboration Manager using the menu bar instead.

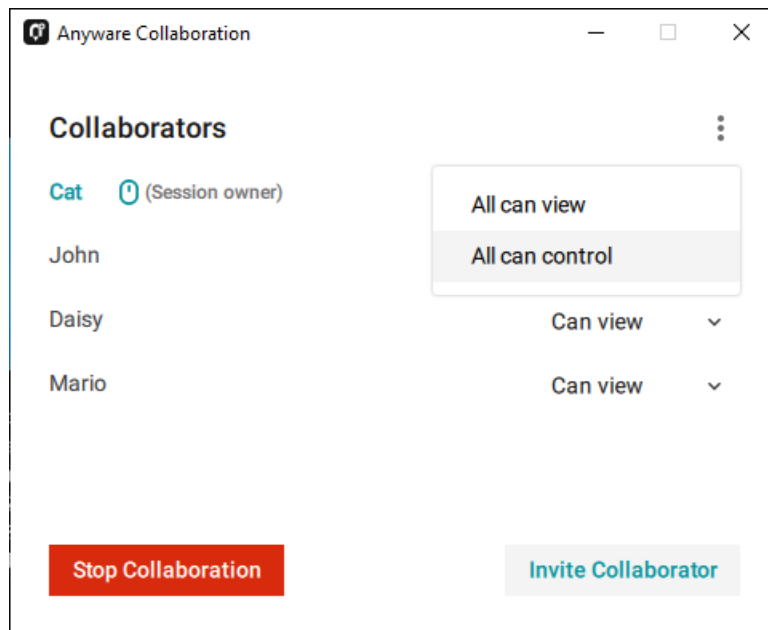
- **From the menu bar:** by clicking the **Anyware Collaboration** icon in the menu bar:



This example shows a Linux desktop; yours may vary depending on which operating system you are connecting to.

2. Grant input control using one of these methods:

- To allow input control for *all collaborators*, click the dropdown menu at the top right and select **All can control**.



- To allow input for *one collaborator*, click the dropdown menu beside the collaborator's name and select **Can control**.

Stopping input control

To return a user (or all users) to view-only mode:

1. From an active collaboration session, open the collaboration manager.
2. Beside the collaborator's name, click the dropdown menu and select **Can View**.

Understanding Display Behavior

Collaboration sessions support sharing of multiple monitors, which varies by session owner's client setting as follows:

- **Windowed mode:** The session in the owner's client window will be shared.
- **Fullscreen One Monitor:** The single fullscreen session will be shared. The session owner should set their Anyware Client to *Fullscreen One Monitor* mode prior to starting the collaboration session.
- **Fullscreen All Monitors:** All monitors will be shared, beginning with the Session Owner's monitor 1 and continuing up to the number of displays in the collaborator's system. The monitors that are shared cannot be configured, and are shared in system order.

*When using this mode, the session owner should assume that collaborators can see **all** displays unless a specific configuration has been tested and verified.*

For example, if the session owner has four monitors and a collaborator only has two, the collaborator will see the session owner's first and second monitors. If different collaborators have a different number of screens, each will see as many displays their system supports; in this scenario, you may have some displays that are visible to certain collaborators but not others.

The session owner should set their Anyware Client to *Fullscreen All Monitors* mode prior to starting the collaboration session.

If the session owner's and collaborator's screen resolutions are different, the collaborator's screen will use scrollbars and letterboxing to display the shared content.

If *high performance client* mode is enabled, and if the session owner's resolution is greater than the collaborator's, the collaborator's screen will be clipped instead.

Joining a Collaboration Session

Collaborators receive the invitations generated by session owners, and use the invitation URI and passcode to connect to the session.

⚠ Important: Anyware Client steps

Collaborators join sessions using Anyware clients. Make sure the software client version you are using supports the collaboration features you require. For details, see [Feature Version Requirements](#).

Each collaborator can join the session with the collaboration link and the Collaboration Invitation passcode. The same URI and passcode are used for all collaborators on the same session.

To join a collaboration session as a collaborator:

1. Open a web browser and go to the collaboration link shared with you (you may be able to click this link directly, depending on how it was shared with you).
2. The web browser will warn you that the link is attempting to open the *Anyware Client* application. Allow the browser to open the Anyware client.
3. When the Anyware client opens, it will prompt you for your name and the Collaboration Invitation passcode. The name you provide here will identify you in the collaboration session. The collaboration invitation passcode is the six digit number provided by the session owner. Enter both values and click **Submit**.
4. Once the session owner accepts your connection request, the Collaboration screen share will start.
5. To leave the collaboration session, select **Connection > Disconnect** from the Anyware Client menu.

Collaborator Input Control

If the session owner has enabled input control for a collaborator, the collaborator can take control of the session owner's desktop including mouse, keyboard, and pointer activity. The session owner retains the ability to stop input control at any time.

Using Input Control as a Collaborator

A collaborator who has input control can release it by idling—stopping all keyboard, mouse, and pointer activity—for a short time. Once the control timeout has elapsed, the floor is open, and whichever collaborator provides input next takes control.

By default, the control timeout is 3 seconds. The timeout value can be configured when enabling the input control feature.

For example: the session owner has initial control of the session. In order to give control to the collaborator, the owner takes their hands off the keyboard and mouse for three seconds, allowing the control timeout to pass. A collaborator then moves their mouse, which gives them control. To give control back to the session owner, the collaborator takes their hands off their keyboard and mouse for three seconds. This exchange continues as long as needed.

Supported Displays

The Graphics Agent for Linux, in combination with a GPU, supports a maximum of four displays on the Anyware client and a maximum resolution of 4K UHD (3840×2160).

Important: Display acceleration requires a supported GPU

The Graphics Agent for Linux must be deployed with a [supported GPU](#). GPUs may have their own licensing requirements.

Anyware agents support any of these monitor configurations:

- Vertical line
- Horizontal line
- Box display

Monitors can be used in any standard rotation (0°, 90°, 180°, or 270°). Any monitor can be the primary.

The Graphics Agent for Linux, in combination with a properly configured *and licensed* GPU, provides the following benefits:

- **Display resolutions:** The Graphics Agent for Linux can provide any resolution a client asks for up to 4K UHD. The practical number of displays available at any given resolution is determined by the GPU, host system hardware, GPU profile licensing, and the connecting client's version.
- **Hot-pluggable displays from Tera2 Anyware Zero Clients:** Displays may be added and removed as needed during an active PCoIP session when using a Tera2 Anyware Zero Client.
- **3D application support:** Full-screen 3D applications are supported.

Note: Using multiple high-resolution displays

Systems with multiple high-resolution displays, such as quad 4K UHD topologies, require powerful system infrastructure. Be sure to use a system with sufficient bandwidth, client and host capabilities, and an appropriate number of licensed GPUs to support your required display topology.

Important: Attaching monitors to the host machine is not supported

Anyware client supports a maximum of four displays. Attaching extra monitors to the host machine will conflict with client display topologies.

Note: A Note on Monitor Orientation

- Monitors with portrait orientation are not supported.
- Monitors with landscape orientation are supported.

PCoIP Ultra

The Graphics Agent for Linux provides support for PCoIP Ultra. PCoIP Ultra is optimized for truly lossless support with bit-exact color accuracy and preservation of content detail at the highest frame rates.

PCoIP Ultra protocol enhancements propels our industry-recognized performance into the future of remote computing, with faster, more interactive experience for users of remote workstations working with high-resolution content.

PCoIP Ultra now defaults to "Auto Offload" on Graphics agent machines, provided that both the client machines and the agent machines are capable of supporting CPU Offload as well as GPU Offload. Additionally, **YUV Chroma subsampling defaults to 4:2:0**. This ensures higher framerates of graphics and optimized motion content, while ensuring efficient utilization of bandwidth.

Note:

PCoIP Ultra defaults to disabled for Standard Agent for Linux.

For additional detail on PCoIP Ultra technical requirements for various use cases and troubleshooting steps, refer to [KB 2109: PCoIP Ultra Troubleshooting](#).

About PCoIP Ultra

PCoIP Ultra is appropriate for users with the following requirements:

Note: Setting the Value of PCoIP Ultra

PCoIP Ultra is disabled by default. You can disable it when recommended based on workload and network conditions.

Auto Offload: Achieves the best balance between color accuracy and network efficiency. This setting is appropriate for work-from-home or WAN content creators who require optimized delivery of high resolution content, including video playback, while still achieving build-to-lossless color accuracy.

CPU Offload: Appropriate for users that require CPU-optimized delivery of 4K UHD, high-framerate video playback and build-to-lossless color accuracy. It is also useful when GPU encoding resources must be reserved for video encoding applications, typically in LAN environments.

GPU Offload: PCoIP encoding is always offloaded to a GPU. Appropriate for users who demand the highest possible CPU efficiency.

Requirements

To take advantage of PCoIP Ultra, you need:

- An **Anyware agent** (any supported version)
- An **Anyware Software Client** (any supported version)

Anyware Tera2 Zero Clients do not support PCoIP Ultra

PCoIP Ultra is only available through Anyware Software Clients and Trusted Zero Clients..

- **CPU offload** requires CPU support for the AVX2 instruction set on both the agent and client machines.
- **GPU offload** requires an NVIDIA graphics card that supports NVENC on the agent machine.

Configuring PCoIP Ultra

To enable PCoIP Ultra features, edit the `pcoip-agent.conf` file and set the `pcoip.ultra` configuration setting as required:

- **1: Turn on *PCoIP Ultra CPU Offload*.**
CPU offload requires CPU support for the AVX2 instruction set on both the remote host and client. The Anyware Zero client is not supported. CPU offload is recommended for 4K UHD resolutions with video playback requirements of 30 fps (or more), and the highest possible image quality and color accuracy.
- **2: Turn on *PCoIP Ultra GPU Offload*.**
GPU Offload is recommended when the CPU impact of pixel encoding should be minimized.
- **3 Turn on *PCoIP Ultra Auto Offload*.**
This setting allows PCoIP to automatically switch between CPU and GPU Offload modes; CPU offload is used by default to provide the best image fidelity, and GPU offload is used during periods of high display activity to provide improved frame rates and bandwidth optimization.

This setting is only effective if the remote host and client endpoints are capable of both CPU and GPU Offload.

All PCoIP Ultra settings take effect on the next PCoIP session. No configuration is required on the Anyware Software Client.

Turning PCoIP Ultra off

To disable PCoIP Ultra, set `pcoip.ultra` to 0.

Setting configuration values

For more information on setting configuration values, refer to [Configuring the Graphics Agent for Linux](#).

Printing Support

The Graphics Agent for Linux does not support printing from local printers connected to Anyware clients. Similarly, printing from USB printers that are connected by means of the USB Bridging feature is also not supported.

The following printing options are available:

- Linux agents can print to any printer on the host agents' local area network.
- If your agent machine has access to the Internet, cloud-based printing is supported through cloud-printing services such as Google Cloud Print and HP Mobile Printing.

USB

USB Support

The Graphics Agent for Linux provides support for USB devices and [certain Wacom tablets](#) attached to Anyware clients.

⚠ USB bridging must be explicitly installed

When installing the Graphics Agent for Linux, you must explicitly enable support for USB bridging by installing the required USB dependencies yourself. Refer to the installation steps for [Ubuntu](#) and [RHEL or Rocky Linux](#) for the required commands. If the required USB dependencies are not installed, the Graphics Agent for Linux will be incapable of bridging USB devices.

This requirement does not affect support for keyboards, and mice or other pointer devices. It does not affect Wacom tablet support.

If the required USB packages are installed, USB bridging support is enabled by default. Administrators can disable or configure USB behavior by changing [configuration options](#).

Keyboards, mice, and other pointer devices are managed by Anyware clients, and are always allowed.

⚠ Important: PCoIP USB allowed and unallowed device rules

The number of PCoIP USB allowed and unallowed device rules in a PCoIP session is determined by the Anyware Client as well as the Anyware Agent. Sessions from Anyware Clients version 23.01 and earlier, as well as sessions from Zero Clients will only apply the first 10 rules.

Xbox One Controller Support

The Graphics Agent for Linux supports Xbox One controllers when attached to Zero Clients **only**.

The following Xbox One controllers are supported:

- Xbox One 2015
- Xbox One
- Xbox One S
- Xbox One Bt
- Xbox One Elite

✎ Support for Xbox Controllers

Xbox Controllers are **currently not supported** by Anyware Software Clients and Trusted Zero Clients.

Tangent Panel Support

The following Tangent panels are supported when connecting from a Windows or Linux software client to a Windows or Linux agent (both 23.04 or higher).

- Tangent Ripple
- Tangent Wave
- Tangent Element BT
- Tangent Element MF
- Tangent Element KB
- Tangent Element TK
- Tangent Arc (Navigation)
- Tangent Arc (Grading)
- Tangent Arc (Function)

macOS Graphics Agents, macOS Software Clients, and Trusted Zero Clients do not support Tangent panels.

Wacom Tablet Support

The Graphics Agent for Linux supports Wacom tablets in two configurations: [bridged](#), where peripheral data is sent to the desktop for processing, and [locally terminated](#), where peripheral data is processed locally at the Anyware client.

Info

When USB auto-forwarding is enabled, Wacom tablets will be forwarded to the host automatically.

Locally Terminated Wacom Tablets

Locally-terminated tablets have greatly improved responsiveness, and tolerate higher-latency (including 25ms and higher) networks.

For the best experience and most complete device support, use the latest available Anyware agent, Anyware software client, and Anyware Zero Client firmware. To find out when support was added for individual Wacom device, refer to the release notes for your client.

⚠ Caution: Using Wacom Local Termination on Ubuntu Cloud Hosts

Cloud-based Ubuntu hosts may fail to properly handle locally terminated Wacom tablets. When this occurs, pressure sensitivity and other advanced features will not work properly. To correct this issue, follow [this procedure](#).

The following Wacom tablet models have been tested and are supported with local termination mode:

Anyware client support for *locally terminated* Wacom tablets and the Graphics Agent for Linux

	Anyware Tera2 Zero Client	Anyware Software Client for Windows	Anyware Software Client for macOS	Anyware Software Client for Linux	Anyware Trusted Zero Client
Intuos Pro Small <i>PTH-460</i>	—	✓	✓	✓	✓
Intuos Pro Medium <i>PTH-660</i>	✓ ¹	✓	✓	✓	✓
Intuos Pro Large <i>PTH-860</i>	✓ ²	✓	✓	✓	—
Intous Pro Small <i>PTK-470</i>	—	✓	✓	✓	—
Intous Pro Medium <i>PTK-670</i>	—	✓	✓	✓	✓
Intous Pro Large <i>PTK-870</i>	—	✓	✓	✓	✓
Movink 13 <i>DTH-135</i>	—	✓ ³	✓ ⁴	✓ ⁵	✓
Cintia 16	—	✓	✓	✓	✓

<i>DTK-168</i>	Anyware Tera2 Zero Client	Anyware Software Client for Windows	Anyware Software Client for macOS	Anyware Software Client for Linux	Anyware Trusted Zero Client
Cintiq 16 (previous generation) <i>DTK-1660</i>	—	✓	✓	✓	✓
Cintiq Pro 16 <i>DTH-167</i>	—	✓	✓	✓	✓
Cintiq Pro 17 <i>DTH-172</i>	—	✓	✓	✓	✓
Cintiq Pro 22 <i>DTH-227</i>	—	✓	✓	✓	✓
Cintiq 24 <i>DTK-246</i>	—	✓	✓	✓	✓
Cintiq 24T <i>DTH-246</i>	—	✓	✓	✓	✓
Cintiq 22 <i>DTK-2260</i>	—	✓	✓	✓	✓
Cintiq 22HD <i>DTK-2200</i>	✓ ⁶	✓	—	✓	✓
Cintiq 22HDT - Pen & Touch <i>DTH-2200</i>	✓ ⁷	—	—	—	—
Cintiq Pro 24 <i>DTK-2420</i>	✓ ⁸	✓	✓	✓	✓
Cintiq Pro 24 - Pen & Touch <i>DTH-2420</i>	✓ ⁹	✓	✓	✓	✓
Cintiq Pro 27 <i>DTH-271</i>	—	✓	✓	✓	✓
Cintiq 32 Pro - Pen & Touch <i>DTH-3220</i>	✓ ¹⁰	✓	✓	✓	✓

⚠ Important: Touch is not supported

Touch features of Wacom devices are not supported with local termination.

- Untested, but expected to work.

Other Wacom tablets may work, but have not been tested and should not be used in production environments.

✎ Info

To ensure smooth and consistent drawing on Wacom tablets in high latency networks, see the [Configuration Guide](#).

Bridged Wacom Tablets

Bridged Wacom tablets are supported only in low-latency environments. Tablets in network environments with greater than 25ms latency will show reduced responsiveness and are not recommended.

The following Wacom tablet models have been tested and are supported with bridged mode:

Anyware client support for *bridged* Wacom tablets and the Graphics Agent for Linux

	Anyware Tera2 Zero Client	Anyware Software Client for Windows	Anyware Software Client for macOS	Anyware Software Client for Linux	Anyware Trusted Zero Client
Intuos Pro Small <i>PTH-460</i>	✓	✓	✓	✓	✓
Intuos Pro Medium <i>PTH-660</i>	✓	✓	✓	✓	✓
Intuos Pro Large <i>PTH-860</i>	✓	✓	✓	✓	✓
Intuos Pro Small <i>PTK-470</i>	—	✓	✓	✓	✓
Intuos Pro Medium <i>PTK-670</i>	—	✓	✓	✓	✓
Intuos Pro Large <i>PTK-870</i>	—	✓	✓	✓	✓
Movink 13 <i>DTH-135</i>	—	✓ ¹¹	✓ ¹²	✓ ¹³	✓ ¹⁴
Cintiq 16 <i>DTK-168</i>	—	✓	✓	✓	✓
Cintiq 16 (previous generation) <i>DTK-1660</i>	—	✓	✓	✓	✓
Cintiq Pro 16 <i>DTH-167</i>	—	✓	✓	✓	✓
Cintiq Pro 17 <i>DTH-172</i>	—	✓	✓	✓	✓
Cintiq Pro 22 <i>DTH-227</i>	—	✓	✓	✓	✓
Cintiq 24 <i>DTK-246</i>	—	✓	✓	✓	✓
Cintiq 24T <i>DTH-246</i>	—	✓	✓	✓	✓
Cintiq 22 <i>DTK-2260</i>	✓	✓	✓	✓	✓
Cintiq 22HD <i>DTK-2200</i>	✓	✓	✓	✓	✓

Cintiq 22HDT - Pen & Touch DTH-2200	Anyware Tera2 Zero Client	Anyware Software Client for Windows	Anyware Software Client for macOS	Anyware Software Client for Linux	Anyware Trusted Zero Client
Cintiq Pro 24 DTK-2420	✓				
Cintiq Pro 24 - Pen & Touch DTH-2420	✓	✓	✓	✓	✓
Cintiq Pro 27 DTH-271	—	✓	✓	✓	✓
Cintiq 32 Pro - Pen & Touch DTH-3220	✓	✓	✓	✓	✓

Note: Support for Other Wacom Tablets

Other Wacom tablets have not been tested, but may work.

Caution: Do Not Calibrate Pen Displays from Wacom Center

Do **NOT** calibrate your pen display from the Wacom Center on the host machine. Doing this can cause cursor offset issues when the tablet is active during a PCoIP session.

- 1 Local termination for Intuos Pro Small and Intuos Pro Medium requires Tera2 Zero Client firmware 6.2.0 or higher.
- 2 Local termination for Intuos Pro Small and Intuos Pro Medium requires Tera2 Zero Client firmware 6.2.0 or higher.
- 3 Untested, but expected to work.
- 4 Untested, but expected to work.
- 5 Untested, but expected to work.
- 6 Local termination for Cintiq 22HD, 22HDT, 24P, and 24PT requires Tera2 Zero Client firmware 6.5.0 or higher.
- 7 Local termination for Cintiq 22HD, 22HDT, 24P, and 24PT requires Tera2 Zero Client firmware 6.5.0 or higher.
- 8 Local termination for Cintiq 22HD, 22HDT, 24P, and 24PT requires Tera2 Zero Client firmware 6.5.0 or higher.
- 9 Local termination for Cintiq 22HD, 22HDT, 24P, and 24PT requires Tera2 Zero Client firmware 6.5.0 or higher.
- 10 Local termination for Cintiq Pro 32PT requires Tera2 Zero Client firmware 20.04 or higher.
- 11 Untested, but expected to work.
- 12 Untested, but expected to work.
- 13 Untested, but expected to work.
- 14 Untested, but expected to work.

Smart Card Authentication

Support for Smart Card Authentication

Smart card authentication is supported when connecting Linux Clients, Windows Clients, or Trusted Zero clients to Linux agents. The following section contains information on system requirements, limitations, agent setup, and client setup.

General Requirements

Linux Distribution	• Ubuntu 22.04 • Rocky 9 • RHEL 9
Client	• Anyware Windows Client 26.01+ • Anyware Linux Client installed on Ubuntu 22.04, version 24.03+ • Anyware Trusted Zero Client 25.03+
Agent	Connections from Windows clients • Graphics Agent for Linux 26.01+ • Standard Agent for Linux 26.01+ Connections from Linux clients and Trusted Zero clients • Graphics Agent for Linux 25.03+ • Standard Agent for Linux 25.03+
Infrastructure	<i>(Required for brokered connections only)</i> • Connection Manager & Security Gateway 20.07+ • Leostream broker

Smart Card Certificate Requirements

The smart card certificate prerequisites are as follows:

- Key usage is set to digital signature
- The Subject common name and subject alternative name (other name) are defined
- Enhanced key usage must include client authentication and/or smart card logon
- Key length does not exceed 2048 bit

Tested Smart Card Readers

The following smart card readers have been tested:

- Belkin USB Smart Card Reader (F1DN008U)

- Identiv SCR3310 USB Contact Smart Card Reader

Tested Smart Card Models

The following smart card models have been tested:

Product Name	Type of Card	Notes
Gemalto TOP DL V2.1 144K FIPS	CAC	
IDEMIA Cosmo v8.0	Alternate token	
IDEMIA ID-one 125 V8.0D	CAC	
G+D Sm@rtCafe Expert v7.0	CAC	
G+D Sm@rtCafe Expert v7.0 144K DI	CAC	
PIVkey C910	PIV	
PIVkey C980	PIV	
PIVkey C990	PIV	
Yubikey 5C		Using PIV interface.
Yubikey 5NFC		Using PIV interface.

Note: Testing Smart Card Solutions

Solutions must be validated in user environments first, as environmental differences including **HP Business Smartcard Keyboard** network conditions or other components may impact support.

Notes

- Smart Card authentication is enabled by modifying the `pcoip.enable_smart_card` directive, as described in [Enabling Smartcard Authentication](#).
- At present, only simultaneous configuration of a single card and single reader is supported.
- Smartcard authentication is only supported on **Ubuntu 22.04 clients** and **Trusted Zero Clients** connecting to Linux agents. It is not supported while connecting from Tera2 Zero Clients to Linux agents.
- If available, configure the Linux Desktop Environment to use smart cards for lock screen authentication.

Known Limitations

- Elliptic Curve Cryptography (ECC) Certificates are not supported.
- Concurrent users cannot log on to agent machines using the same smart card for authentication.
- Smart cards having multiple certificates allow only one user to log on at a time. Other users must wait until the current users logs off before attempting to log on.

- Single sign-on is not supported. Users must authenticate twice: once on the client and again on the host machine's lock screen.
- Session locking upon smart card removal might not work as expected.
- Session will disconnect when authenticated smart card is removed from the reader.
- If PCoIP sessions fail, disable SELINUX and re-establish a PCoIP session. If this does not work, contact the [HP Support team](#).
- If PIN prompts do not work on Linux Desktop Environments of agent machines, use the password to unlock the desktop. Smartcards will be remoted and available to be used in session.

Agent Setup

Note

Some card readers might require their drivers to be installed on the agent machine. Consult with the reader manual to determine whether you need to install the required drivers.

Prerequisites

- The host machine is domain-joined.
- **If this is a brokered connection**, make sure that you installed Leostream broker and Connection Manager.
- CA certificates for authenticating smart cards are available.

Step I: Prepare the Linux Machine

1. Connecting the machine directly to AD using SSSD. For more information, consult the following topics:
 - [Connecting RHEL systems directly to AD using SSSD](#)
 - [Set up SSSD with Active Directory on Ubuntu machines](#)
2. Enable smart card authentication on the Linux machine and configure smart card for lock screen. For more information, consult the following topics:
 - [Configuring smart card authentication on RHEL machines](#)
 - [Configuring smart card authentication on Ubuntu](#)
3. Install the Leostream agent on the Linux machine. For more information, see the [Leostream® Platform Installation Guide](#).

Info

For common errors encountered during Leostream agent configuration, see the [Leostream Install Errors article](#).

Step II: Install and Configure Anyware Agent

1. Make sure that you downloaded Anyware Agent 25.03 or later to the remote machine.
2. Install the agent following instructions in [Installing the agent](#)
3. Enable smart card authentication on the agent:

1. Navigate to `/etc/pcoip/`.
2. Open the `pcoip-agent.conf` file.
3. Locate the `pcoip.enable_smart_card` directive.
4. Set its value to "1".
5. Save your changes.
6. Add root and intermediate CA certificates to the agent's trusted certificate store:
 - **Ubuntu** - using `update-ca-certificates`, see [Install a PEM-format certificate](#).
 - **RHEL/Rocky** - using `update-ca-trust`, see [Adding new certificates](#).
7. Add the following setting to the domain section in the **sssd.conf** file, available at the following location:
`/etc/sss/`

For example:

```
[domain/example.net]
ad_gpo_map_permit = +pcoip-session
```

- Domain section
- In a multiple card scenario, to make sure SSSD picks up the correct card/certificate for authentication, the following setting can be added to `/etc/sss/sss.conf`

1. Open the sssd config file:

```
sudo vi /etc/sss/sss.conf
```

2. Add the following setting to the [pam] section `p11_uri = pkcs11:token=[your-token-name]`
3. Restart sssd service:

```
sudo systemctl restart sssd
```

For RHEL/Rocky a system reboot might be needed for this to work.

Note

To determine the token label or other attributes to use, you can run the following command:

```
p11tool --list-token-urls
```

8. Restart the SSSD service by running the following command:

```
systemctl restart pcsd
```

Note

The following steps **9** to **17** are only applicable for **Ubuntu 22.04**.

9. Ensure that smart card is showing in the lock screen. Create a user file in the `/etc/dconf/profile` .

```
cat << EOF | sudo tee -a /etc/dconf/profile/user
user-db:user
system-db:local
EOF
```

10. Create the `local.d` directory.

```
sudo mkdir -p /etc/dconf/db/local.d
```

11. Disable the password authentication.

```
cat << EOF | sudo tee -a /etc/dconf/db/local.d/00_gdm-password-disable
[ora/anome/login-screen]
enable-password-authentication=false
EOF
```

12. Enable the smart card authentication.

```
cat << EOF | sudo tee -a /etc/dconf/db/local.d/00_gdm-smartcard-enable
[ora/anome/login-screen]
enable-smartcard-authentication=true
EOF
```

13. Create the **locks** directory.

```
sudo mkdir -p /etc/dconf/db/local.d/locks
```

14. Make **enable-password-authentication** and **enable-smartcard-authentication** non-user configurable. Add the following lines to the locks directory to prevent users from changing these settings:

```
cat << EOF | sudo tee -a /etc/dconf/db/local.d/locks/00_gdm-smartcard-locks
```

15. Make these options non-user configurable:

```
/ora/anome/login-screen/enable-password-authentication
/ora/anome/login-screen/enable-smartcard-authentication
EOF
```

16. To update the **system dconf database** after making changes, run the following command:

```
sudo dconf update
```

17. Restart the machine.

```
sudo reboot
```

Client Setup

1. Download the Anyware client:

- Anyware Linux Client version 25.03 or later on an Ubuntu 22.04 client machine
 - Anyware Windows client version 26.01 or later on a Windows machine
 - Anyware Trusted Zero client version 25.03 or later
2. Configure the client machine to connect to the agent machine. Follow the instructions in the topic "Connecting to an Agent Machine " in the Anyware Client guide.
 3. Plug the smart card reader into the client machine.
 4. Use your smart card to authenticate the session. For a full set of instructions on using smart cards to authenticate PCoIP sessions, consult "Using Smart Card Authentication to Connect to a Session" in the Anyware Client guide.

Configuring Smartcard Removal Behavior

This policy determines the action to be taken when the smart card used to authenticate the session is removed from the card reader, or the card reader is disconnected.

Info

If the smart card removal behavior is not configured, removing the card will disconnect the session.

1. On the agent machine, navigate to `/etc/pcoip/` .
2. Open the `pcoip-agent.conf` file.
3. Set the `smart_card_removal_behavior` directive to configure the smart card removal behavior:
 - To disconnect session on smart card removal (default behavior), set `pcoip.smart_card_removal_behavior` to `"1"`.
 - To take no action on removal of smart card, set `pcoip.smart_card_removal_behavior` to `"0"`.
4. Restart the system.

Disabling Smart Card Support

1. On the agent machine, navigate to `/etc/pcoip/` .
2. Open the `pcoip-agent.conf` file.
3. Set the `pcoip.enable_smart_card` directive to `"0"`.
4. Save your changes.
5. Reboot the agent machine.

Troubleshooting Issues

For higher latency networks where the latency between the client and host is higher than 100 milliseconds, authentication using smart cards on lock screen may fail.

This occurs because an SSSD component called `p11_child` times out before the pin gets validated. To workaround this issue, increase the `SSSD p11_child` time-out by making these changes in the `/etc/sss/sss.conf` file:

1. In the `[pam]` section, increase the `p11_child_timeout` value to 60 seconds.

2. Restart SSSD service.

```
systemctl restart sssd.service
```

Reporting Issues

To report issues with this feature, enable additional debugging capabilities before creating a support bundle as below:

1. On the host machine, uncomment the following line from the `/etc/default/pcoip-pcscd-args` file: `PCSCD_ARGS=--debug`
2. Save your changes.
3. Reproduce the issue.
4. Generate support bundle and send it to the [HP Anyware support team](#).

Info

An example procedure for setting up and enabling smartcard authentication on Linux machines is available in the [Reference section](#).

Installation Guide

Installation Overview

The following pages describe the overall installation process for both RHEL/Rocky Linux and Ubuntu systems.

⚠ Important: A Desktop environment is required

Before you begin, ensure a desktop environment is installed. Any desktop environment will work; the following options are provided as a convenience, but you can install any available environment.

- **RHEL or Rocky Linux:**

```
sudo yum groupinstall 'Server with GUI'
```

- **Ubuntu:**

Kubuntu distributions are bundled with KDE; you can install KDE from other distributions by using this command:

```
sudo apt install kubuntu-desktop
```

To install Mate Desktop, use this command:

```
sudo apt install ubuntu-mate-desktop
```

Before You Begin

Confirm the following items before starting the installation process:

- The Graphics Agent for Linux has specific requirements for GPU acceleration. Make sure your machine is configured with a [supported card](#) before you start.
- SSH must be enabled.
- You must have a license registration code for the agent instance from HP (as part of an HP Anyware subscription).
- The desktop machine requires the following ports to be open: TCP 443, TCP 60443, TCP 4172, and UDP 4172.
- Your GPUs must be licensed with their manufacturers (if required).
- You must have super user (root) privileges and be able to issue `sudo` commands.
- If you are using a Local License Server, you'll need to know its URL and port numbers.

⚠ Important: Protect your license registration code

The license registration code you receive from HP is unique to your organization, and should be protected as you would any sensitive data.

Be careful that you do not inadvertently expose your registration code in forums or other public areas by pasting log messages without redacting sensitive information.

Installation Overview

Once your prerequisites are in place, you can proceed with installation. Here's a brief overview of the process:

1. Connect to the machine using SSH.
2. Install the appropriate [GPU drivers](#) for your system.
3. Install the Anyware Agent on a [Anyware Agent](#).
4. If required, [configure](#) the agent software.
5. Disconnect the SSH session.
6. Connect to the desktop using a Anyware client.

If you're ready to start, connect to your machine with an SSH client and proceed to [Installing GPU Drivers](#).

Installing GPU Drivers

Installing GPU Drivers on Linux machines

Before installing the Anyware Agent, install the correct driver for your configured GPUs. SSH into your machine and then proceed with the installation instructions below.

The correct driver versions for release 26.05 are shown below. Be sure to install the specified driver, and *not the latest available version*. We qualify Graphics Agent support against the driver versions listed here.

Documentation for download and installation of these drivers is provided by their manufacturers. As a convenience, we've provided the current locations of relevant third-party documentation here. While we do our best to keep these links up to date, we do not control these resources or their locations and it's possible that they could break.

⚠ RHEL/Rocky Linux Only: Install `libglvnd-opengl` before proceeding

When using **RHEL OR Rocky Linux**, install the `libglvnd-opengl` package before installing the GPU drivers:

```
sudo yum install libglvnd-opengl
```

Install NVIDIA GRID Drivers on AWS Instances

Instance Type	Supported GPUs	Supported NVIDIA GRID driver version	Installation instructions
EC2 G4dn	NVIDIA T4	GRID 19.4+	
Download the driver from Amazon using the <i>AWS CLI</i> or <i>SDKs</i> , as documented by Amazon .			
G4 instances must also include NVIDIA licensing, as documented by NVIDIA .			
EC2 G5	NVIDIA A10G	GRID 19.4+	
Download the driver from Amazon using the <i>AWS CLI</i> or <i>SDKs</i> , as documented by Amazon .			
G5 instances must also include NVIDIA licensing, as documented by NVIDIA .			

Install NVIDIA Drivers on Physical PCs

Instance Type	Supported GPUs	Supported NVIDIA driver version	Installation instructions
Physical PC	• Tesla (any) • Quadro (k2000+)	Use the latest 580.126.09+ driver for your supported GPU	Download the driver directly from NVIDIA and follow their instructions to install. You may only have access to this driver if you have purchased a corresponding card.

Install NVIDIA GRID Drivers on Azure NV-series Instances

Instance Type	Supported GPUs	Supported NVIDIA GRID driver version	Installation instructions
Azure NV-series	NVIDIA M60, T4	GRID 18.6+	Microsoft Azure download

Install NVIDIA GRID Drivers on Other Instance Types

Instance Type	Supported GPUs	Supported NVIDIA GRID driver version	Installation instructions
All other instance types	GPUs as listed in NVIDIA's GRID release notes .	GRID 19.4+	Instructions shown below.

Installing NVIDIA Drivers on All Other Instances

For all other instance types, including non-cloud instances, use the following procedures:

Install NVIDIA Drivers on RHEL or Rocky Linux

⚠ Install `libglvnd-opengl` before proceeding

Before installing the NVIDIA drivers, install the `libglvnd-opengl` package:

```
sudo yum install libglvnd-opengl
```

1. Install tools required to install the NVIDIA drivers:

```
sudo yum install kernel-devel
```

2. Disable the Nouveau video driver (two commands):

```
echo 'blacklist nouveau' | sudo tee -a /etc/modprobe.d/blacklist.conf
```

3. As a super user (sudo), edit the file `/etc/default/grub`. Append the following to the **GRUB_CMDLINE_LINUX** entry:

```
rd.driver.blacklist=nouveau nouveau.modeset=0
```

For example:

```
GRUB_CMDLINE_LINUX="crashkernel=auto console=ttyS0,38400n8 rd.driver.blacklist=nouveau nouveau.modeset=0"
```

Generate a new grub configuration to include the above changes, then reboot:

```
sudo grub2-mkconfig -o /boot/grub2/grub.cfg  
sudo reboot
```

4. Install the NVIDIA driver:

```
sudo ./NVIDIA-Linux-x86_64-xxx.xx-grid.run --dkms
```

Where `xxx.xx` is the NVIDIA driver version shown in Linux System Requirements.

⚠ Important: Azure installations require Hyper-V daemon installation

If you are running on an Azure instance, you also need to install the Azure Hyper-V daemon:

```
sudo yum install hyperv-daemons
```

5. Respond to the installer prompts as follows:

- Accept the EULA
- Say *yes* to installing 32-bit binaries
- Say *no* to modifying the x.org file

6. Update initramfs using dracut:

```
dracut -fv
```

7. Verify the NVIDIA installation:

```
nvidia-smi
```

If the installation was successful, you will see your video card in the response.

Install NVIDIA Drivers on Ubuntu

1. Install tools required to install the NVIDIA drivers:

```
sudo apt install build-essential
```

2. Disable the Nouveau video driver (two commands):

```
echo 'blacklist nouveau' | sudo tee -a /etc/modprobe.d/blacklist.conf
```

3. As a super user (sudo), edit the file `/etc/default/grub`. Append the following to the **GRUB_CMDLINE_LINUX** entry:

```
rd.driver.blacklist=nouveau nouveau.modeset=0
```

For example:

```
GRUB_CMDLINE_LINUX="crashkernel=auto console=ttyS0,38400n8 rd.driver.blacklist=nouveau nouveau.modeset=0"
```

Generate a new grub configuration to include the above changes, then reboot:

```
sudo grub2-mkconfig -o /boot/grub2/grub.cfg  
sudo reboot
```

4. Install the NVIDIA driver:

```
sudo ./NVIDIA-Linux-x86_64-xxx.xx-grid.run --dkms
```

Where `xxx.xx` is the NVIDIA driver version shown in Linux System Requirements.

5. Respond to the installer prompts as follows:

- Accept the EULA
- Say *yes* to installing 32-bit binaries
- Say *no* to modifying the x.org file

6. Verify the NVIDIA installation:

```
nvidia-smi
```

If the installation was successful, you will see your video card in the response.

Installing the Anyware Agent

Once the NVIDIA drivers are present, you can install the [Graphics Agent for Linux](#).

Installing the Agent

Instructions are provided here for installing the Graphics Agent for Linux on [RHEL/Rocky Linux](#) and [Ubuntu](#). If you are installing on a physical machine, see the special instructions [provided below](#).

Note: Disabling GSP

Some GPUs include a GPU System Processor (GSP), which may be used to offload GPU initialization and management tasks. GSP is supported only for vCS in GPU pass through and bare-metal deployments. If you are using any other product in a GPU pass through or bare-metal deployment, you must disable the GSP firmware. Failure to disable GSP can lead to inability in resizing desktops or incorrect display configuration. For more information, see this article on [the NVIDIA Docs Site](#).

Important: Required ports will be automatically opened

The Graphics Agent for Linux installer will add firewall exceptions for the following required PCoIP ports during installation: TCP 443, TCP 4172, UDP 4172, and TCP 60443.

Note: Desktop user interfaces will only be available using PCoIP

Once installed and running, the Anyware Graphics Agent for Linux takes over the graphics subsystem which is then unavailable to hypervisors. You can only view the graphical user interface when connecting with a Anyware client.

For example, you cannot view an ESXi virtual machine console through VSphere; you must connect to the machine using PCoIP.

Installing the Agent on RHEL or Rocky Linux

These instructions are for RHEL or Rocky Linux machines. For Ubuntu instructions, see [Installing the Agent on Ubuntu](#), below.

Important: IP Configuration Considerations

For the Anyware Agent installation to complete successfully, the RHEL/Rocky 8 host machine must be configured to use both IPv4 and IPv6.

To install the Anyware Graphics Agent for Linux software on a RHEL/Rocky Linux machine:

1. Download and install the repository via the [shell script provided here](#).
2. Install the EPEL repository:

```
sudo yum install epel-release
```

3. **Optionally** install USB dependencies, if you intend to support USB devices other than keyboards, mice, and pointer devices. **If you skip this step, USB redirection will be completely disabled and bridged USB devices will not work.**

```
sudo yum install usb-vhci
```

4. Install the Anyware Graphics Agent for Linux:

```
sudo yum install pcoip-agent-graphics
```

5. Note your machine's local IP address. Clients connecting directly to the host workstation will need this number to connect.
6. Enter the license registration code you received from us.

 Note: These instructions are for Cloud Licensing

These instructions assume you are using Anyware Cloud Licensing to activate your PCoIP session licenses. If you are using the License Server instead, see [Licensing the Graphics Agent for Linux](#).

For unproxied internet connections, type:

```
pcoip-register-host --registration-code=<XXXXXX@YYY-YYYY-YYY>
```

For proxied internet connections, type:

```
pcoip-register-host --registration-code=<XXXXXX@YYY-YYYY-YYY> --proxy-server=<serverURL> --proxy-port=<port>
```

7. Reboot the desktop.

GNOME Display Manager Support

GNOME Display Manager (GDM) is supported by the Graphics Agent for Linux as a credential authenticator when gdm-runtime-config is available. This allows PCoIP sessions to be locked and unlocked within the remoted session. When the PCoIP service starts, GDM is configured to run without obstructing access to the GPU. This configuration is reverted when the PCoIP service stops. This behavior is always on and requires no configuration on the Anyware agent. For more information on GDM, see the [GNOME Display Manager Reference Manual](#).

Installing the Agent on Ubuntu

These instructions are for Ubuntu machines. For RHEL/Rocky Linux instructions, see [Installing the Agent on RHEL or Rocky Linux](#), above.

To install the Anyware Graphics Agent for Linux software on an Ubuntu machine:

1. Download and install the repository via the [shell script provided here](#).
2. **Optionally** install USB dependencies, if you intend to support USB devices other than keyboards, mice, and pointer devices. *If you skip this step, USB redirection will be completely disabled and bridged USB devices will not work.*

```
sudo apt install usb-vhci-dkms
```

3. Install the Anyware Graphics Agent for Linux:

```
sudo apt update  
sudo apt install pcoip-agent-graphics
```

4. Note your machine's local IP address. Clients connecting directly to the host workstation will need this number to connect.
5. Enter the license registration code you received from us.

 Note: These instructions are for Cloud Licensing

These instructions assume you are using Anyware Cloud Licensing to activate your PCoIP session licenses. If you are using the License Server instead, see [Licensing the Graphics Agent for Linux](#).

For unproxied internet connections, type:

```
pcoip-register-host --registration-code=<XXXXXX@YYY-YYYY-YYY>
```

For proxied internet connections, type:

```
pcoip-register-host --registration-code=<XXXXXX@YYY-YYYY-YYY> --proxy-server=<serverURL> --proxy-port=<port>
```

6. Reboot the desktop.

 Note: Desktop user interfaces will only be available using PCoIP

Once installed and running, the Anyware Graphics Agent for Linux takes over the graphics subsystem which is then unavailable to hypervisors. You can only view the graphical user interface when connecting with a Anyware client.

For example, you cannot view an ESXi virtual machine console through VSphere; you must connect to the machine using PCoIP.

Installing USB Drivers on Secure Boot Enabled Linux Machines

If Secure Boot is enabled on a Linux host machine with UEFI, USB drivers cannot be installed on that host machine. This stops the client-side USB devices from getting bridged to the host machine. To prevent this from occurring, perform the following steps, provided that **you have access to the UEFI Firmware Menu**.

 Note: If You Do Not Have UEFI Firmware Menu Access

If you do not have access to the UEFI Firmware Menu, either disable Secure Boot or deploy the Virtual Machine without Secure Boot.

1. Connect to the remote host machine via SSH.
2. Run the following command:

```
sudo mokutil --import /var/lib/dkms/mok.pub
```

3. If the Machine Owner Key (MOK) is not set, the mokutil utility will prompt you to create one.
4. Reboot the machine.
5. Enroll the MOK in the UEFI firmware menu.
6. Reboot the machine again.

7. Run the following commands:

```
lsmod | grep usb  
dmesg | grep vhci
```

The output of this command indicates that `usb-vhci` is installed on the host machine.

Installing on a Physical Machine

When installing the Anyware Agent on a physical machine with a Quadro card, additional steps are required. **If you installed the Graphics Agent for Linux on a virtual machine, you can ignore these steps.**

Note: Install the agent first

You must install the Graphics Agent for Linux as [described above](#) before proceeding.

Important: Some GPUs are automatically configured

The Graphics Agent for Linux can use the following NVIDIA GPUs automatically:

- K2000
- K2200
- K4000
- K4200
- K5200
- M4000
- P2000
- P4000
- P5000
- P6000
- RTX4000

If you are unable to start a session using any of these GPUs, follow the procedure described next.

These steps are only required when installing on a physical, non-virtualized machine with a Quadro card:

1. Attempt to start a PCoIP session. **If the connection attempt succeeds, you can stop here.** Your GPU will work without further configuration. **If the attempt fails**, continue on; we will use the log information generated by the failed attempt in the next steps.
2. Look in `/var/log/Xorg.100.log` for lines like this:

```
[ 293.834] (--) NVIDIA(GPU-0): LNX Linux XGA (DFP-0): connected
[ 293.834] (--) NVIDIA(GPU-0): LNX Linux XGA (DFP-0): Internal DisplayPort
[ 293.834] (--) NVIDIA(GPU-0): LNX Linux XGA (DFP-0): 1440.0 MHz maximum pixel clock
[ 293.834] (--) NVIDIA(GPU-0):
[ 293.834] (--) NVIDIA(GPU-0): LNX Linux XGA (DFP-1): connected
[ 293.834] (--) NVIDIA(GPU-0): LNX Linux XGA (DFP-1): Internal TMDS
[ 293.834] (--) NVIDIA(GPU-0): LNX Linux XGA (DFP-1): 165.0 MHz maximum pixel clock
[ 293.834] (--) NVIDIA(GPU-0):
[ 293.834] (--) NVIDIA(GPU-0): LNX Linux XGA (DFP-2): connected
[ 293.834] (--) NVIDIA(GPU-0): LNX Linux XGA (DFP-2): Internal DisplayPort
[ 293.834] (--) NVIDIA(GPU-0): LNX Linux XGA (DFP-2): 1440.0 MHz maximum pixel clock
[ 293.835] (--) NVIDIA(GPU-0):
[ 293.835] (--) NVIDIA(GPU-0): LNX Linux XGA (DFP-3): connected
[ 293.835] (--) NVIDIA(GPU-0): LNX Linux XGA (DFP-3): Internal TMDS
[ 293.835] (--) NVIDIA(GPU-0): LNX Linux XGA (DFP-3): 165.0 MHz maximum pixel clock
[ 293.835] (--) NVIDIA(GPU-0):
[ 293.835] (--) NVIDIA(GPU-0): DFP-4: disconnected
[ 293.835] (--) NVIDIA(GPU-0): DFP-4: Internal DisplayPort
[ 293.835] (--) NVIDIA(GPU-0): DFP-4: 1440.0 MHz maximum pixel clock
[ 293.835] (--) NVIDIA(GPU-0):
[ 293.835] (--) NVIDIA(GPU-0): DFP-5: disconnected
[ 293.835] (--) NVIDIA(GPU-0): DFP-5: Internal TMDS
[ 293.835] (--) NVIDIA(GPU-0): DFP-5: 165.0 MHz maximum pixel clock
[ 293.835] (--) NVIDIA(GPU-0):
[ 293.835] (--) NVIDIA(GPU-0): DFP-6: disconnected
[ 293.835] (--) NVIDIA(GPU-0): DFP-6: Internal DisplayPort
[ 293.835] (--) NVIDIA(GPU-0): DFP-6: 1440.0 MHz maximum pixel clock
[ 293.835] (--) NVIDIA(GPU-0):
[ 293.835] (--) NVIDIA(GPU-0): DFP-7: disconnected
[ 293.835] (--) NVIDIA(GPU-0): DFP-7: Internal TMDS
[ 293.835] (--) NVIDIA(GPU-0): DFP-7: 165.0 MHz maximum pixel clock
```

- Note the names of the outputs with the highest maximum pixel clocks (as many as four will be shown). In the above example, you would note `DFP-0` , `DFP-2` , `DFP-4` , and `DFP-6` .

 Note: Systems with fewer than four displays (Quadro K series)

If there are fewer than four outputs in your system, note them all. K-series cards only support two enabled outputs at a time, but may show more.

- Create a file called `/etc/X11/xorg.conf.d/10-pcoip.conf`:

```
touch /etc/X11/xorg.conf.d/10-pcoip.conf
```

- Open `/etc/X11/xorg.conf.d/10-pcoip.conf` in a text editor.
- Paste the following text into the file:

```
Section "Screen"
Identifier "dummy_screen"
Device "dummy_videocard"
Option "UseDisplayDevice" ""
Option "ConnectedMonitor" ""
Option "Monitor-" "Monitor0"
Option "Monitor-" "Monitor1"
Option "Monitor-" "Monitor2"
Option "Monitor-" "Monitor3"
Monitor "Monitor0"
EndSection
```

- Populate the `ConnectedMonitor` and `UseDisplayDevice` settings with the display IDs you noted earlier, as a comma-separated list. Using our example (your values may be different):

```
Option "UseDisplayDevice" "DFP-0,DFP-2,DFP-4,DFP-6"  
Option "ConnectedMonitor" "DFP-0,DFP-2,DFP-4,DFP-6"
```

8. Next, map these output values to your expected displays. Modify the monitor options to correspond to the displays in the previous step. Each output should be mapped to a monitor identifier from one of the `Monitor` sections in `10-pcoip.conf`.

Continuing our example, we would map the four outputs to four monitors like this (yours may be different; for example, you may only have two outputs):

```
Option "Monitor-DFP-0" "Monitor0"  
Option "Monitor-DFP-2" "Monitor1"  
Option "Monitor-DFP-4" "Monitor2"  
Option "Monitor-DFP-6" "Monitor3"
```

9. Save and close `10-pcoip.conf`.
10. Attempt another PCoIP session. If the session is still unsuccessful, and you have a Quadro K series card, repeat steps 7 and 8 with a different output combination, e.g. if you tried `DFP-3,DFP-4`, try `DFP-1,DFP-2`. Due to driver differences certain output combinations may not work.
11. To prevent the NVIDIA driver from changing the kernel mode setting, add the `nomodeset` parameter to the GRUB configuration:
 1. Open `/etc/default/grub` in a text editor.
 2. Add `nomodeset` to `GRUB_CMDLINE_LINUX_DEFAULT`.
 3. Save and close the file.
 4. Rebuild the GRUB configuration:

- **RHEL/Rocky Linux**

```
sudo update-grub
```

- **Ubuntu**

```
sudo grub2-mkconfig -o /boot/grub2/grub.cfg
```

5. Reboot the machine.

12. Attempt a PCoIP connection again. The connection should succeed.

If the session does not start, make sure that the displays are correctly configured.

Once you've installed the software, you can [configure it](#), [register licenses](#), or [connect to it](#).

Installing the Agent in Offline Environment

Installing the Agent Offline

Users can now install the Linux agents even if they are offline. Follow the step by step instructions given below.

Steps to Install Linux on the Target Anyware Agent Machine in Offline Environment:

1. On your **Agent** machine, navigate to **HP Anyware Website**.

2. Download `tar.gz` file from HP website.

Example: `pcoip-agent-offline-rocky9.7_26.05.0-1.el9.x86_64.tar.gz`

1. After downloading the file, extract the installation files from `tar.gz` file.

2. Use the below command for extraction `tar.gz` file.

```
sudo tar xvf pcoip-agent-offline-rocky8.6_23.12.6-1.el8.x86_64.tar.gz
```

1. Install the agent by running the following command:

```
sudo ./install-pcoip-agent.sh
```

Licensing the Agent

The Graphics Agent for Linux must be assigned a valid PCoIP session license before it will work. Until you've registered it, you can't connect to the desktop using a Anyware client.

You receive a registration code when you purchase a pool of licenses from HP. Each registration code can be used multiple times; each use consumes one license in its pool.

Note: Registration code format

Registration codes look like this: `ABCDEFGH12@AB12-C345-D67E-89FG`

Anyware agent license registrations are managed automatically by HP's [Cloud Licensing service](#). If necessary, you can manage them yourself, using your own locally-installed [License server](#) instead.

If you need to purchase licenses, contact [HP](#).

Troubleshooting Licensing Issues

If you're encountering problems with HP licensing, refer to [Troubleshooting License Issues](#).

Using HP Cloud Licensing

To use Cloud Licensing, all you need to do is provide a registration code for each Anyware agent in your deployment (the same registration code can be used multiple times).

To provide the registration code:

SSH into the agent machine, and invoke `pcoip-register-host` with the license registration code and proxy settings if required:

```
pcoip-register-host --registration-code=<registration-code> [--proxy-server=<proxy-server-address>] [--proxy-port=<proxy-port-number>]
```

Allowlist Network Blocks for Anyware Cloud Licensing

If you are using Anyware Cloud Licensing, you will need to add the following to your allowlist:

- `teradici.flexnetoperations.com`
- `teradici.compliance.flexnetoperations.com`

If you use an IP-based allowlist, we recommend your IT team add the following network blocks to your allowlist:

- IPv4: `185.146.155.64/27`
- IPv6: `2620:122:f005::/56`

⚠ Important: Migrating from the previous specification

Previously, our allowlist specification looked like this:

- **Production:** 64.14.29.0/24
- **Disaster Recovery:** 64.27.162.0/24

If you have an existing implementation using an IP-based allowlist like this, we recommend you leave it in place until the new allowlist is active and tested.

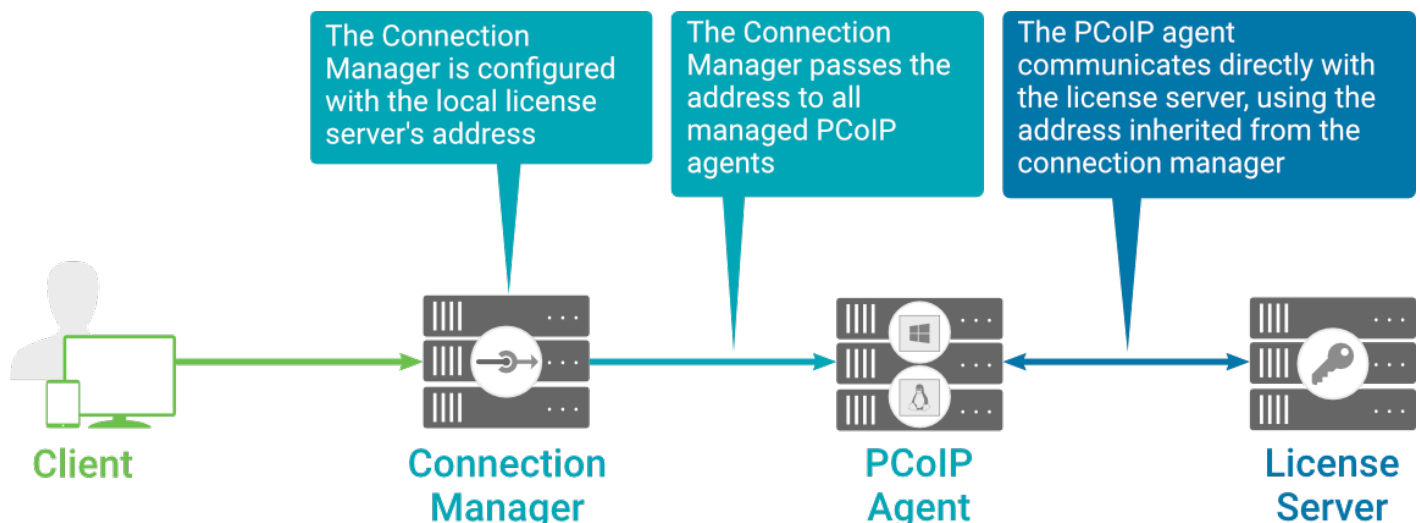
Licensing Anyware Agents With a Local License Server

In deployments where Anyware agents cannot access the internet, or where cloud-based licensing is not permitted or desired, a local License Server can be used instead. The License Server manages PCoIP session licenses within your private environment.

Configuring Anyware agents to use a local license server is done in one of two ways, depending on whether your deployment uses a PCoIP Connection Manager, or whether your Anyware clients connect directly to Anyware agents.

Brokered Environment Licensing

In *brokered* deployments, the license server address is configured in the Connection Manager, which passes it through to its managed Anyware agents.



Local license validation using a Graphics Agent for Linux and a brokered connection

When using a Connection Manager, the license server address is only configured once no matter how many Anyware agents are behind the Connection Manager.

To set the License Server URL in the Connection Manager:

1. On the Connection Manager machine, use a text editor to open `/etc/ConnectionManager.conf`.
2. Set the `LicenseServerAddress` parameter with the address of your local license server:
 - `http:// {license-server-address} : {port} /request`
3. Save and close the configuration file.
4. Restart the Connection Manager.

Verifying Your Brokered Licensing Configuration

To verify your system's licensing configuration, run `pcoip-validate-license` from the console on the Graphics Agent for Linux machine. The command will ping the license server and attempt to retrieve information on an available license:

```
pcoip-validate-license --license-server-url <license-server-address> [--proxy-server <proxy-server-address>] [--proxy-port <proxy-port-number>]
```

Where `<license-server-address>` is the address of the license server to ping, formatted as `http:// {license-server-address} : {port} /request`

If the license server is behind a proxy server, provide the proxy information via the `--proxy-server` and `--proxy-port` parameters.

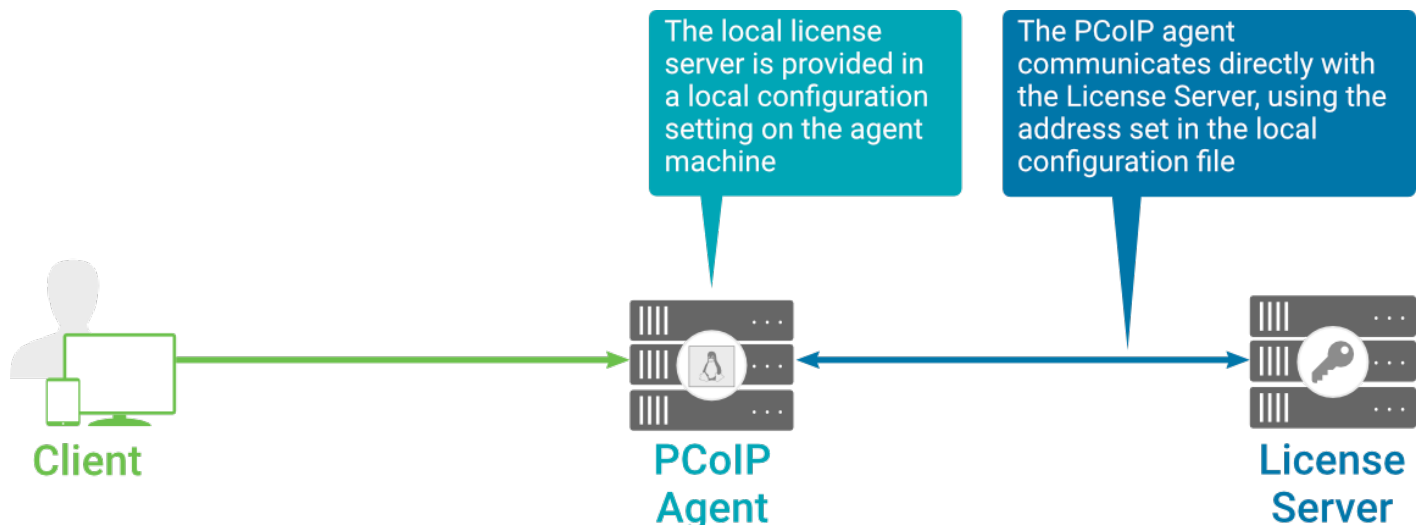
If successful, the response will show that a valid license was found on the license server, and its expiration date.

If the connection is unsuccessful, investigate the following possibilities:

- The license server address is incorrect, or formatted incorrectly.
- The license server is inaccessible.
- There are no available licenses on the license server. `pcoip-validate-license` will only return a positive response if there is at least one available session license.
- If you have only one license on the license server and run `pcoip-validate-license` from a PCoIP session, the command will fail because you are currently using the single license. In this scenario, disconnect your PCoIP session and try again from an SSH session instead.

Unbrokered Environment Licensing

In direct, or unbrokered, deployments, each Anyware agent is configured with the license server address via a local agent setting. When a client initiates a new PCoIP session, the Anyware agent uses its local configuration to communicate with the license server.



Local license validation using a Graphics Agent for Linux and a direct (unbrokered) connection

Each Anyware agent in your environment must be individually configured with the license server's URL.

To configure the License Server URL on the Graphics Agent for Linux machine:

1. Using a text editor, open `/etc/pcoip-agent/pcoip-agent.conf`.
2. Add or modify the `pcoip.license_server_path` directive:

```
pcoip.license_server_path = <license-server-address>
```

Where `<license-server-address>` is the address of the license server, formatted as `http:// {license-server-address} : {port} /request`.

3. If the license server is behind a proxy server, provide the proxy information using the `pcoip.license_proxy_server` and `pcoip.license_proxy_port` directives.
4. Save and close `pcoip-agent.conf`.

The changes will take effect on the next PCoIP session.

Verifying Your Unbrokered Licensing Configuration

To verify your system's licensing configuration, run `pcoip-validate-license` from the console on the Graphics Agent for Linux machine. The command will ping the license server and attempt to retrieve information on an available license:

```
pcoip-validate-license --license-server-url <license-server-address> [--proxy-server <proxy-server-address>] [--proxy-port <proxy-port-number>]
```

Where `<license-server-address>` is the address of the license server to ping, formatted as `http:// {license-server-address} : {port} /request`

If the license server is behind a proxy server, provide the proxy information via the `--proxy-server` and `--proxy-port` parameters.

If successful, the response will show that a valid license was found on the license server, and its expiration date.

If the connection is unsuccessful, investigate the following possibilities:

- The license server address is incorrect, or formatted incorrectly.
- The license server is inaccessible.
- There are no available licenses on the license server. `pcoip-validate-license` will only return a positive response if there is at least one available session license.
- If you have only one license on the license server and run `pcoip-validate-license` from a PCoIP session, the command will fail because you are currently using the single license. In this scenario, disconnect your PCoIP session and try again from an SSH session instead.

Updating the Agent

Updates to the Graphics Agent for Linux will be published on a regular basis. New stable builds will be produced approximately every three months.

To update the Graphics Agent for Linux to the current version, update your local package as described below. Your configuration settings will be preserved.

To update the Graphics Agent for Linux:

- **RHEL/Rocky Linux**

Open a terminal and enter the following commands:

```
sudo yum makecache
sudo yum update pcoip-agent-graphics
sudo reboot
```

- **Ubuntu**

```
sudo apt update
sudo apt install pcoip-agent-graphics
sudo reboot
```

Uninstalling the Agent

You can remove the Graphics Agent for Linux from your system, or you can remove the repo config entirely.

Remove the Graphics Agent for Linux package

To remove the package, open a console window and run the following command:

- **RHEL/Rocky Linux**

```
sudo yum remove pcoip-agent-*
```

- **Ubuntu**

```
sudo apt-get remove pcoip-agent-*
```

Remove the repo configuration

You can also remove the repo configuration completely. You'll need to do this if you are switching from one channel to another (for example, from beta to stable), before reconfiguring with the new repo:

Note: Removing beta repo configurations

The following commands remove the *stable* channel, which is what most users have configured. To remove the *beta* channel instead, replace `pcoip-agent` with `pcoip-agent-beta` in these commands.

- **RHEL/Rocky Linux**

```
rm /etc/yum.repos.d/pcoip-agent.repo  
rm /etc/yum.repos.d/pcoip-agent-source.repo
```

- **Ubuntu**

```
rm /etc/apt/sources.list.d/pcoip-agent.list  
apt-get clean  
rm -rf /var/lib/apt/lists/*  
apt-get update
```

Configuration Guide

Configuring the Anyware Agent

You can configure the Anyware agent, and optimize PCoIP protocol behavior for local network conditions, by adjusting configuration directives found in `/etc/pcoip-agent/pcoip-agent.conf`.

You can find detailed information and descriptions about each setting [in the next section](#). You can also consult the `man` pages for `pcoip-agent.conf`:

```
man pcoip-agent.conf
```

Applying Configuration Changes

To set or change a configuration value, add or modify directives in `pcoip-agent.conf`. Place one directive on each line, in this format:

```
directive.name = <value>
```

For example, to set the [maximum frame rate](#) to *60 frames per second*, set the [maximum bandwidth](#) to *900000 kilobits/second*, and the [device bandwidth floor](#) to *5000 kilobits/second*, you would set values in `pcoip-agent.conf` like this:

```
pcoip.maximum_frame_rate = 60
pcoip.max_link_rate = 900000
pcoip.device_bandwidth_floor = 5000
```

Support for IPv6 Addresses

The Graphics Agent for Linux supports IPv6 addresses. No configuration is needed to switch between IPv4 and IPv6 modes.

H.264 Hardware Decode Requirements

For H.264 Hardware Decode, the **Graphics Agent** must have an NVIDIA Graphics Card that supports PCoIP Ultra GPU Offload, and PCoIP Ultra setting is either set to GPU Offload or Auto Offload.

Default Values

In deployments with no existing PCoIP Ultra configuration, PCoIP Ultra defaults to "Auto Offload" and YUV Chroma subsampling defaults to 4:2:0.

The following NVIDIA graphics cards are supported:

- NVIDIA Quadro P400
- NVIDIA GeForce RTX 3060

Any Nvidia GPU with NVENC support are expected to work, but have not been tested.

Configurable Values

A complete list of configurable values is shown next in [Configurable Settings](#)

Configurable Settings

The following settings can be configured on the Graphics Agent for Linux. Refer to [Configuring the Anyware agent](#) to understand how to modify these settings.

Authentication broker URL

Directive	Options	Default
pcoip.authentication_broker_url	string (<i>up to 511 characters</i>)	—

This setting takes effect when you start the next session. This policy sets the authentication broker URL for federated user authentication. Enter the authentication broker URL in 'https://address:port/auth' format. This setting overwrites the Authentication broker URL from Connection Manager.

Build-to-lossless

Directive	Options	Default
pcoip.enable_build_to_lossless	0 (off), 1 (on)	Off

This setting takes effect immediately. Specifies whether to turn the build-to-lossless feature of the PCoIP protocol off or on; this feature is turned off by default.

When build-to-lossless is turned off images and other desktop content may never build to a lossless state. In network environments with constrained bandwidth, turning off build-to-lossless can provide bandwidth savings. Build-to-lossless is recommended for environments that require images and desktop content to be built to a lossless state.

Clipboard settings

Directive	Options	Range	Increment	Default
pcoip.server_clipboard_state	0—Disabled in both directions 1—Enabled in both directions 2—Enabled client to agent only 3—Enabled agent to client only			—
pcoip.max_clipboard_data_size		0 – 20971520	1	—
pcoip.clipboard_content_type	0—All 1—Text only 2—Image only			—

Clipboard Redirection: This setting takes effect when you start the next session. Determines the direction in which clipboard redirection is allowed. You can select one of these values:

- Disabled in both directions
- Enabled in both directions (default setting)
- Enabled client to agent only (That is, allow copy and paste only from the client system to the host desktop.)
- Enabled agent to client only (That is, allow copy and paste only from the host desktop to the client system.)

Clipboard redirection is implemented as a virtual channel. If virtual channels are disabled, clipboard redirection does not function.

When this setting is disabled or not configured, the default value is Enabled in both directions.

Maximum Clipboard Data Size: This setting takes effect when you start the next session. It specifies the maximum clipboard size in bytes, with 0 using the default maximum. The maximum value allowed is 20971520 bytes (20 MB). When plain text exceeds the configured limit, it will be truncated at or below the limit to preserve character boundaries.

When this setting is disabled or not configured, the default value of 0 is used.

Allowed Clipboard Content Types: This setting takes effect when you start the next session. Controls what types of content can be copied and pasted through the clipboard redirection feature. You can select one of these values:

- All (default setting)
- Text only
- Image only

When this setting is disabled or not configured, the default value is All content types.

Collaboration

Directive	Options	Range	Increment	Default
pcoip.enable_collaboration	0 (off), 1 (on)			Off
pcoip.max_collaborators		1 – 5	1	5
pcoip.collaboration_udpport		1 – 65535	1	64172

This setting takes effect when the agent is restarted. This policy enables or disables user collaboration. When not configured, user collaboration is disabled by default.

The default maximum number of collaborators allowed is 5.

The default UDP starting port used for collaborator sessions is 64172. When a different starting port is used, ensure that firewall rules are adjusted so that PCoIP traffic can go through the new port.

If there is more than one collaborator, additional UDP ports will be needed for the collaborator sessions. For example, when the second collaborator connects, the next free UDP port will be opened on the host.

WARNING! Collaboration is not currently supported when connecting with Anyware Manager Enterprise, including HP Anyware for Windows 365. If you enable collaboration with Anyware Manager Enterprise your session will run in a lower performance mode.

Collaboration input control

Directive	Options	Range	Increment	Default
pcoip.enable_collaboration_input_control	0 (off), 1 (on)			Off
pcoip.collaboration_input_control_timeout		100 – 10000	100	3000

This setting takes effect when you start the next PCoIP session. This policy enables or disables input control from the collaborators. When not configured, collaboration input control is disabled by default.

The input control timeout specifies the waiting period before any user with input control permission can acquire the input control of the host. The current input owner is the only one authorized to send mouse, keyboard and touch inputs to the host.

Connection addresses

Directive	Options	Default
pcoip.connection_address	string (up to 511 characters)	—
pcoip.client_connection_address	string (up to 511 characters)	—

This setting takes effect when you start the next session. Configuring this allows you to control the IPv4 or IPv6 address used by the agent or client in PCoIP sessions.

'Connection Address' controls the IP address used by the agent for the PCoIP session.

'Client Connection Address' controls the IP address the client is told to use when establishing the PCoIP session.

Please note that neither of these values should need to be set under normal circumstances.

Desktop environment

Directive	Options	Default
pcoip.desktop_session	string (up to 511 characters)	—

This setting takes effect when you start the next session after rebooting or logging out of the desktop. Choose the desktop environment that will be launched from `/usr/share/xsessions/*.desktop`. If no desktop is specified, the system will automatically select the first available desktop in the following order:

- Ubuntu
- GNOME
- Plasma
- MATE

If none of these desktops are present, the agent will use the first desktop found alphabetically in `/usr/share/xsessions/`.

Note that this setting only takes effect after an existing desktop session ends, either due to a reboot or logging out. Simply disconnecting a PCoIP session will not log a user out of the desktop session.

Important: Specify only the filename WITHOUT the `.desktop` extension. For example, to use the MATE desktop (which corresponds to `/usr/share/xsessions/mate.desktop`), set: `pcoip.desktop_session = mate`

Enable Disclaimer Authentication

Directive	Options	Default
pcoip.enable_disclaimer_auth	0 (off), 1 (on)	Off

This setting takes effect when you start the next session. When this setting is enabled, users connecting via direct connect will be presented a disclaimer prior to user authentication. If the disclaimer is rejected, the user will not be able to connect.

Disclaimer files must be placed in */etc/pcoip-agent/disclaimers/* and must be readable by the "pcoip" system user. Files must be named according to the locale, e.g. en_US.txt for en_US, ko_KR.txt for ko_KR, etc. If a file matching the negotiated locale is not present, en_US will be used as a fallback. If disclaimer text cannot be found, an blank disclaimer will be presented.

Enable smart card support

Directive	Options	Default
pcoip.enable_smart_card	0—disable (default) 1—enable smart card authentication and in-session use	—

This setting takes effect when you start the next session. Smart card PIN must be used for this feature. When enabled, the smart card can be used for session authentication, lock screen login, document signing, and other in-session smart card uses. When not configured, smart card authentication and in-session use will be disabled by default.

Enable/disable USB in the PCoIP session

Directive	Options	Default
pcoip.enable_usb	0 (off), 1 (on)	On

This setting takes effect when you start the next session. Determines whether USB support is enabled in PCoIP sessions. When this setting is not configured, USB is enabled by default. By default all devices are supported unless restrictions are configured through the USB device rules setting.

Enable/disable audio in the PCoIP session

Directive	Options	Default
pcoip.enable_audio	0 (off), 1 (on)	On

This setting takes effect when you start the next session. Determines whether audio is enabled in PCoIP sessions. Both endpoints must have audio enabled. When this setting is enabled, PCoIP audio is allowed. When it is disabled, PCoIP audio is disabled. When this setting is not configured, audio is enabled by default.

Enable/disable relative mouse support

Directive	Options	Default
pcoip.enable_relative_mouse	0 (off), 1 (on)	On

This setting takes effect when you start the next session. It determines whether relative mouse co-ordinates may be used,

when appropriate, during the PCoIP session. By default, this setting is enabled.

Hide local cursor

Directive	Options	Default
<code>pcoip.disable_locally_rendered_cursor</code>	0 (off), 1 (on)	Off

This setting takes effect immediately. When this setting is enabled the local cursor on the client will be hidden. This may resolve duplicate cursor issues if there is a host rendered cursor within the host environment but may also result in no visible cursor. With this setting enabled there may be delays in mouse movements due to network latency and video processing times. By default, this setting is disabled, meaning that local cursors will be used, providing the most responsive user experience.

Hide remote cursor

Directive	Options	Default
<code>pcoip.disable_server_rendered_cursor</code>	0 (off), 1 (on)	On

This setting takes effect after a session reconnection. When this setting is enabled, the remote cursor on the server will be hidden. By default, the local cursor is rendered while the remote cursor is not for a better user experience. With this setting set to disabled, there may be delays in mouse movements due to network latency and video processing times. When both host side and client side cursors are reporting as visible, the client side cursor will automatically be hidden.

Host key auto repeats

Directive	Options	Default
<code>pcoip.use_host_autorepeat</code>	0 —disabled (default) 1 —use host generated auto repeats and client repeat rate (zero client only) 2 —use host generated auto repeats and host repeat rate	—

This setting takes effect when you start the next session. Configuring this allows you to enable or disable host generated key auto repeats. When not configured or disabled, key auto repeats are driven by the client.

Improve graphics tablet responsiveness on high latency network

Directive	Options	Default
<code>pcoip.allow_lossy_hid_reports</code>	0 (off), 1 (on)	On

This setting takes effect when you start the next session.

When this setting is enabled and connected with a compatible client, performance and responsiveness on the locally terminated Wacom/Xencelabs graphics tablet will be improved on networks with high latency or occasional packet drops, though there may be a slight reduction in accuracy.

This setting is not applicable to USB bridged graphics tablets.

By default, this setting is enabled.

License server URL

Directive	Options	Default
pcoip.license_server_path	string (up to 511 characters)	—

This setting takes effect when you start the next session. This policy sets the license server path. Enter the license server path in 'https://address:port/request' or 'http://address:port/request' format.

Maximum PCoIP session bandwidth

Directive	Range	Increment	Default
pcoip.max_link_rate	104 – 900000	100	900000

This setting takes effect when you start the next session. Specifies the maximum bandwidth, in kilobits per second, in a PCoIP session. The bandwidth includes all imaging, audio, virtual channel, USB, and control PCoIP traffic.

Set this value based on the overall capacity of the link to which your endpoint is connected, taking into consideration the number of expected concurrent PCoIP sessions. For example, with a single user VDI configuration (e.g. a single PCoIP session) that connects through a 4Mbit/s Internet connection, set this value to 4Mbit (or 10% less than this value to leave some allowance for other network traffic).

Setting this value prevents the agent from attempting to transmit at a higher rate than the link capacity, which would cause excessive packet loss and a poorer user experience. This value is symmetric. It forces the client and agent to use the lower of the two values that are set on the client and agent side. For example, setting a 4Mbit/s maximum bandwidth forces the agent to transmit at a lower rate, even though the setting is configured on the client.

When this setting is disabled or not configured on an endpoint, the endpoint imposes no bandwidth constraints. When this setting is configured, the setting is used as the endpoint's maximum bandwidth constraint in kilobits per second.

The default value when this setting is not configured is 900000 kilobits per second.

This setting applies to the agent and client. If the two endpoints have different settings, the lower value is used.

PCoIP Security Certificate Settings

Directive	Options	Default
pcoip.ssl_cert_type	1 —From certificate storage 2 —Generate a unique self-signed certificate 0 —From certificate storage if possible, otherwise generate	—
pcoip.ssl_cert_min_key_length	1024 —1024 bits 2048 —2048 bits 3072 —3072 bits 4096 —4096 bits	—

This setting takes effect when you start the next session. A certificate is used to secure PCoIP related communications. The way PCoIP components choose a certificate is based on the certificate type and the key length. Without a certificate being generated or selected, a PCoIP Session cannot be established.

Depending on the value chosen for the option, 'How the PCoIP agent chooses the certificate...' and the availability of appropriate certificates, PCoIP components may acquire a CA signed certificate from certificate storage or generate an in-memory self-signed certificate.

In order for a CA signed certificate to be loadable by PCoIP components, it must be stored at `/etc/pcoip-agent/ssl-certs` in

three .pem files, owned by the pcoip user, only readable by the owning user.

- pcoip-key.pem must contain an unlocked RSA key
- pcoip-cert.pem must contain a certificate that signs the key in pcoip.pem
- pcoip-cacert.pem must contain a CA certificate chain that validates the certificate in pcoip-cert.pem.

Note: Self-signed certificates are 3072 bits long.

Select a minimum key length (in bits) for a CA signed certificate. Longer length certificates will require more computing resources and may reduce performance, but will increase security. Shorter length certificates will provide better performance at the cost of lower security.

Note: Please refer to Teradici documentation for instructions on creating and deploying certificates.

PCoIP Security Settings

Directive	Options	Default
pcoip.tls_cipher_blacklist	string (up to 1023 characters)	—

This setting controls the cryptographic cipher suites used by PCoIP endpoints. Changes will take effect when the agent is restarted. When this setting is disabled or not configured, all supported cipher suites may be used for connections. The endpoints negotiate the actual cryptographic cipher suites based on the settings configured here. Newer versions of TLS and stronger cipher suites will be preferred during negotiation between endpoints. Supported cipher suites:

- TLS_ECDHE_RSA_WITH_AES_256_GCM_SHA384
- TLS_DHE_RSA_WITH_AES_256_GCM_SHA384
- TLS_AES_256_GCM_SHA384

Blacklisted Cipher Suites Provides the ability to block specific cipher suites from being offered during negotiation. Must be entered as a semi-colon separated list of cipher suites.

PCoIP USB allowed and unallowed device rules

Directive	Options	Default
pcoip.usb_auth_table	23XXXXXX 2203XXXX	23XXXXXX
pcoip.usb_unauth_table	2203XXXX	—

This setting takes effect when you start the next session. This setting only applies if the related setting to Enable/disable USB in the PCoIP session is enabled, and specifies the USB devices that are authorized and not authorized for PCoIP sessions. Only devices listed in the USB authorization table are permitted in PCoIP sessions, provided they are not subsequently excluded by an entry in the USB unauthorization table.

You can define a maximum of 50 USB authorization rules and a maximum of 50 USB unauthorization rules. Separate multiple rules with the vertical bar (|) character. Please note the final number of authorization/unauthorization rules in a PCoIP session are negotiated by PCoIP client and agent. Some clients have a limit of 10 USB rules. Please refer to the PCoIP agent admin guide for details.

Each rule can be a combination of a Vendor ID (VID) and a Product ID (PID), or a rule can describe a class of USB devices. A class rule can allow or disallow an entire device class, a single subclass, or a protocol within a subclass.

The format of a combination VID/PID rule is 1xxxxyyyy, where xxxx is the VID in hexadecimal format and yyyy is the PID in hexadecimal format. For example, the rule to authorize or block a device with VID 0x1a2b and PID 0x3c4d is 11a2b3c4d.

For class rules, use one of the following formats:

Allow all USB Format: 23XXXXXX devices Example: 23XXXXXX

Allow USB Format: 22classXXXX devices with a Example: 22aaXXXX specific class ID

Allow a specific Format: 21class-subclassXX subclass Example: 21aabbXX

Allow a specific Format: 20class-subclass-protocol protocol Example: 20aabbcc

For example, the USB authorization string to allow USB HID (mouse and keyboard) devices (class ID 0x03) and mass storage devices (class ID 0x08) is 2203XXXX|2208XXXX. The USB unauthorization string to disallow USB Mass Storage devices (class ID 0x08) is 2208XXXX.

An empty USB authorization string means that no USB devices are authorized. An empty USB unauthorization string means that only USB devices in the authorization list are allowed.

If these settings are unconfigured, the default behavior is that all devices are allowed.

PCoIP Ultra

Directive	Options	Range	Increment	Default
pcoip.enable_ultra	0 (off), 1 (on)			On
pcoip.ultra	0—Disabled 1—CPU Offload 2—GPU Offload 3—Automatic Offload			—
pcoip.ultra_offload_mpps		1 – 40	1	10

This setting takes effect when you start the next session. When this setting is disabled, PCoIP Ultra will not be used.

- PCoIP Ultra CPU Offload - these optimizations require CPU support for the AVX2 instruction set on both the remote host and client and are not compatible with the PCoIP Zero client. CPU Offload is recommended for 4K UHD resolutions with video playback requirements of 30 fps (or more) and highest image quality / color accuracy.
- PCoIP Ultra GPU Offload - these optimizations require an NVIDIA graphics card on the remote host capable of NVENC. GPU Offload is recommended when minimal CPU impact of pixel encoding is desired.
- PCoIP Ultra Auto Offload - enabling this setting allows PCoIP to automatically switch between CPU and GPU Offload modes; CPU Offload is used by default to provide the best image fidelity, GPU Offload is used during periods of high display activity to provide improved frame rates and bandwidth optimization. This setting is only effective if the remote host and client endpoints are capable of both CPU and GPU Offload.

The PCoIP Ultra Offload MPPS sets the Megapixels Per Second (MPPS) transition rate between PCoIP Ultra CPU Offload and PCoIP Ultra GPU Offload. Under Auto-Offload, PCoIP Ultra uses CPU Offload at lower pixel rates and switches to GPU Offload at the Offload MPPS. Increasing this value results in PCoIP Ultra transitioning to GPU Offload at a higher pixel rate and decreasing this value results in the transition at a lower pixel rate. The default PCoIP Ultra Offload MPPS is set to 10.

PCoIP event log verbosity

Directive	Range	Increment	Default
pcoip.event_filter_mode	0 – 3	1	2

This setting takes effect immediately. Configures the PCoIP event log verbosity ranging from 0 (least verbose) to 3 (most verbose).

PCoIP image quality levels

Directive	Options	Range	Increment	Default
pcoip.minimum_image_quality		20 – 100	10	40
pcoip.maximum_initial_image_quality		30 – 100	10	80
pcoip.frame_rate_vs_quality_factor		0 – 100	10	50
pcoip.maximum_frame_rate		0 – 60	1	—
pcoip.yuv_chroma_subsampling	0—4:4:4 1—4:2:0			—
pcoip.use_client_img_settings	0 (off), 1 (on)			Off

This setting takes effect immediately. Controls how PCoIP renders images during periods of network congestion. The Minimum Image Quality, Maximum Initial Image Quality, and Maximum Frame Rate values interoperate to provide fine control in network-bandwidth constrained environments.

Use the Minimum Image Quality value to balance image quality and frame rate for limited-bandwidth scenarios. You can specify a value between 30 and 100. The default value is 40. A lower value allows higher frame-rates, but with a potentially lower quality display. A higher value provides higher image quality, but with potentially lower frame rates when network bandwidth is constrained. When network bandwidth is not constrained, PCoIP maintains maximum quality regardless of this value.

Use the Maximum Initial Image Quality value to reduce the network bandwidth peaks required by PCoIP by limiting the initial quality of the changed regions of the display image. You can specify a value between 30 and 100. The default value is 80. A lower value reduces the image quality of content changes and decreases peak bandwidth requirements. A higher value increases the image quality of content changes and increases peak bandwidth requirements. Unchanged regions of the image progressively build to a lossless (perfect) quality regardless of this value. A value of 80 or lower best utilizes the available bandwidth.

The Minimum Image Quality value cannot exceed the Maximum Initial Image Quality value.

Use the Frame Rate vs Image Quality value to favor image sharpness over smooth motion during a PCoIP session when network bandwidth is limited. Lower values favor smoothness, higher values favor sharpness of image.

Use the Maximum Frame Rate value to manage the average bandwidth consumed per user by limiting the number of screen updates per second. You can specify a value between 1 and 60 frames per second. A higher value can use more bandwidth but provides less jitter, which allows smoother transitions in changing images such as video. A lower value uses less bandwidth but results in more jitter.

YUV chroma subsampling is set to 4:2:0 by default. It enables chroma subsampling to further compress the imaging to reduce bandwidth usage at the cost of reduced color accuracy. 4:2:0 subsampling is only supported in combination with PCoIP Ultra GPU optimization. Please note: 4:4:4 subsampling with PCoIP Ultra GPU optimization is GPU dependent and is not supported by all GPUs, in this case, PCoIP will fall back to 4:2:0 subsampling. Please see our support site for further details.

Set the 'Use image settings from zero client' when you want to use the 'Minimum Image Quality', 'Maximum Initial Image Quality', 'Maximum Frame Rate', 'Disable Build to Lossless' values from the client instead of the host. Currently, only Zero Client Firmware 3.5 and above support these settings on the client side.

These image quality values apply to the soft host only and have no effect on a soft client.

When this setting is disabled or not configured, the default values are used.

PCoIP session MTU

Directive	Range	Increment	Default
pcoip.mtu_size	500 – 1500	1	1200

This setting takes effect when you start the next session. Specifies the Maximum Transmission Unit (MTU) size for UDP packets for a PCoIP session.

The MTU size includes IP and UDP packet headers. TCP uses the standard MTU discovery mechanism to set MTU and is not affected by this setting. The maximum MTU size is 1500 bytes. The minimum MTU size is 500 bytes. The default value is 1200 bytes.

Typically, you do not have to change the MTU size. Change this value if you have an unusual network setup that causes PCoIP packet fragmentation.

This setting applies to the agent and client. If the two endpoints have different MTU size settings, the lowest size is used.

If this setting is disabled or not configured, the client uses the default value in the negotiation with the agent.

PCoIP session audio bandwidth limit

Directive	Range	Increment	Default
pcoip.audio_bandwidth_limit	0 – 100000	1	512

This setting takes effect immediately. Specifies the maximum audio bandwidth that can be used for audio output (sound playback) from the virtual desktop to the client in a PCoIP session. Note that the network transport overhead can add an additional 20-40% bandwidth to this number.

Audio processing monitors the bandwidth needed for audio and selects the audio compression algorithm that provides the best quality possible, without exceeding the bandwidth limit:

- 512 kbit/s or higher - 7.1 surround, high-quality, compressed audio
- 384 kbit/s or higher - 5.1 surround, high-quality, compressed audio
- 256 kbit/s or higher - stereo, high-quality, compressed audio
- 48 kbit/s to 255 kbit/s - stereo audio ranging between FM radio quality down to AM radio quality
- 32 kbit/s to 47 kbit/s - monaural AM radio or phone call quality
- Below 32 kbit/s - results in no audio playback

If this setting is not configured, a default audio bandwidth limit of 512 kbit/s is configured to constrain the audio compression algorithm selected.

Note that zero clients on older firmware have less efficient audio compression algorithms that may require setting this limit higher to achieve the same audio quality or upgrading the firmware.

PCoIP session bandwidth floor

Directive	Range	Increment	Default
pcoip.device_bandwidth_floor	0 – 900000	1	—

This setting takes effect immediately. Specifies a lower limit, in kilobits per second, for the bandwidth that is reserved by the PCoIP session.

This setting configures the minimum expected bandwidth transmission rate for the endpoint. When you use this setting to reserve bandwidth for an endpoint, the session does not have to wait for bandwidth to become available, which improves session responsiveness.

Make sure that you do not over-subscribe the total reserved bandwidth for all endpoints. Make sure that the sum of bandwidth floors for all connections in your configuration does not exceed the network capability.

The default value is 0, which means that no minimum bandwidth is reserved. When this setting is disabled or not configured, no minimum bandwidth is reserved.

This setting applies to the agent and client, but the setting only affects the endpoint on which it is configured.

PCoIP statistics interval

Directive	Range	Increment	Default
<code>pcoip.server_statistics_interval_seconds</code>	0 – 65535	1	—

This setting takes effect immediately. Configuring this allows you to set an interval in seconds for logging performance statistics to the PCoIP server log. When not configured, logging is disabled by default.

PCoIP transport header

Directive	Options	Default
<code>pcoip.transport_session_priority</code>	1 —High Priority 2 —Medium Priority (default) 3 —Low Priority 4 —Undefined Priority	—

This setting takes effect when you start the next session. Configures the PCoIP transport header.

PCoIP transport header is a 32-bit long header which is added to all PCoIP UDP packets (only if the transport header is enabled/supported by both sides). PCoIP transport header allows network devices to make better prioritization/Qos decisions when dealing with network congestions. The transport header is enabled by default.

The transport session priority determines the PCoIP session priority reported in the PCoIP Transport Header. Network devices make better prioritization/Qos decisions based on the specified transport session priority. The transport session priority value is negotiated by the PCoIP agent and client. If agent has specified a transport session priority value (high, medium, or low), then the session uses the agent specified session priority. If only the client has specified a transport session priority (high, medium, or low), then the session uses the client specified session priority. If neither agent nor client has specified a transport session priority (or specified 'undefined priority'), then the session uses/defaults to the medium session priority.

PCoIP virtual channels

Directive	Options	Default
<code>pcoip.enable_vchan</code>	1 —Enable all virtual channels other than those in the list 2 —Disable all virtual channels other than those in the list	—
<code>pcoip.vchan_list</code>	string (<i>up to 255 characters</i>)	—

These settings take effect when you start the next session and specify what virtual channels can operate over a PCoIP session.

There are two modes of operation that can be selected by the "Virtual channel" setting:

- Enabled: Enable all virtual channels except for those in <Virtual channel list> (default setting)
- Disabled: Disable all virtual channels except for those in <Virtual channel list>

For additional security, virtual channel plugin signing can be verified. The "Signature Verification for Virtual Channel Plugins" setting determines how it is enforced:

- Enabled: Only signed virtual channel plugins are allowed. Unsigned plugins will be blocked unless explicitly listed in the exception list.
- Disabled: All virtual channel plugins will be loaded, regardless of whether they are signed. (default setting)

When specifying a virtual channel list, the following rules apply:

- An empty list is allowed
- Multiple virtual channel names in the list must be separated by the vertical bar (|) character. For example: channelA|channelB
- Vertical bar or backslash (\) characters in virtual channel names must be preceded by a backslash. For example: the channel name "awk\ward\channel" must be specified as "awk\\ward\\channel" (without the double quotes)
- A maximum of 15 virtual channels are allowed in a single PCoIP session

The virtual channel must be enabled on both agent and client for it to be used.

Proxy Access to a remote License Server

Directive	Options	Range	Increment	Default
pcoip.license_proxy_server	string (up to 511 characters)			—
pcoip.license_proxy_port		0 – 65535	1	—

This setting takes effect when you start the next session. If a proxy is required to access a local License Server or the Cloud License Server, enter those parameters here. These parameters are loaded only during agent startup.

Session Automatic Reconnection Policy

Directive	Range	Increment	Default
pcoip.session_retry_timeout	0 – 120	1	20

This setting takes effect when you start the next session. This policy configures the automatic reconnection period, that is the amount of time a PCoIP Client and Server will attempt to reconnect an interrupted session without requiring the user to re-enter their logon credentials.

A session may be interrupted through network loss, for instance through pulling a network cable, disabling a network interface or moving away from a WiFi hotspot. In the case of portable computing devices closing a laptop lid or similar actions have the same effect. By default, when network connectivity returns within the specified time period, the session will be restored with no further user action being required.

If this setting is disabled or not configured, the default reconnect period is 20 minutes.

Setting this value to 0 disables the session automatic reconnection feature but allows for session reconnection as a result of intermittent short term network loss (between 30 and 60 seconds).

Smart card removal behavior

Directive	Options	Default
pcoip.smart_card_removal_behavior	0—no action 1—disconnect session (default)	—

This setting takes effect when you start the next session. This policy determines what action is taken when the smart card used to authenticate the session is removed from the card reader, or the card reader is disconnected. Smart card support must be enabled. When not configured, smart card removal will result in a disconnected session by default.

Timezone redirection

Directive	Options	Default
pcoip.enable_timezone_redirect	0 (off), 1 (on)	On

This setting takes effect when you start the next session. Configuring this allows you to enable or disable timezone redirection. When not configured, timezone redirection is enabled by default.

Username comparison skipping

Directive	Options	Default
pcoip.skip_username_comparison	0 (off), 1 (on)	Off

This setting takes effect when the agent is restarted. It allows username comparison to be skipped when launching the user's desktop environment. It should only be enabled when using a PAM stack where the desktop user may differ from the username used during login.

X display number

Directive	Range	Increment	Default
pcoip.x_display_number	0 – 65535	1	100

This setting only takes effect after an existing desktop session ends, either due to a reboot or logging out. This setting sets the X display number used in the PCoIP sessions. Default is 100.

If the display number is already in use and the session cannot be established, try using a different display number.

Simply disconnecting a PCoIP session will not log a user out of the desktop session.

X server remote access

Directive	Options	Default
pcoip.allow_x_remoting	0 (off), 1 (on)	Off

This setting takes effect when you restart the agent. Configuring this allows you to enable or disable remote access to the X server run by the PCoIP Agent. When not configured, remote access is disabled by default.

Connection Guide

Making a Connection from a PCoIP Client

Note: Choose the Appropriate User

While connecting to agent machines, make sure that the username or the user group is not `pcoip` .

Anyware Agent Deployment and Client Connectivity Requirements

Anyware clients can connect to your desktops hosted in proof-of-concept, cloud, or datacenter deployments. Requirements and network security levels will vary depending on your deployment type.

Note: Access to Clipboard Data

If clipboard redirection is enabled, Anyware agents have access to clipboard data. We recommend that you **disable clipboard redirection** if you are handling sensitive data.

Once you've installed and configured your Graphics Agent for Windows, you're ready to accept incoming connections from remote Anyware Clients. PCoIP clients are remote endpoint devices available in as software or firmware and make secure PCoIP connections to the remote desktop through the installed Graphics Agent for Windows.

Managing Client Connections

In most cases, Anyware clients connect to Anyware agents through a connection broker. The broker is responsible for matching users to their available desktops, and then establishing the PCoIP session with their selected resource.

Anyware agents do not need to be configured to use these brokering services. All relevant configuration is done at the broker, which then communicates with the agent.

Brokering Options

There are several ways you can manage client connections to remote desktops

Direct Connections

In direct connection scenarios—where a broker is not involved—the Anyware agent acts as its own broker. In these cases, a client user will provide the IP address or FQDN of the agent machine to their client, and the connection is made securely with no intermediate step.

Anyware Manager

[Anyware Manager](#) is a service, available as a cloud-based service or as an installable instance, that centrally manages PCoIP deployments. It enables highly scalable and cost-effective HP Anyware deployments by managing cloud compute

costs and brokering PCoIP connections to remote Windows or Linux workstations.

Anyware Manager Enterprise

[Anyware Manager Enterprise](#) is an end-to-end SaaS solution for efficient, secure, and streamlined management of HP Anyware deployments. Anyware Manager Enterprise can rapidly deploy virtual desktops, provide remote access to digital workspaces, and provide visibility into the state of digital workspaces. As a unified platform, it enables organizations to manage complex, distributed infrastructures, regardless of deployment model (cloud, on-premises, or hybrid).

Connection Manager

The **Connection Manager** is provided in a bundle with the **Security Gateway**, and is available for use with Third Party brokers. For information about the Connection Manager, including installation and configuration instructions, see the [Connection Manager and Security Gateway documentation](#).

Third-party Connection Brokers

Anyware agents also support third-party connection brokers. For a current list of brokering partners, see [Technology Partners](#) on the website.

Additional Information

Information about **Anyware client connectivity requirements and usage instructions**, is available in the following documentation:

Software clients:

[Anyware Software Client for Windows](#)

[Anyware Software Client for macOS](#)

[Anyware Software Client for Linux](#)

Mobile Clients:

[Anyware Mobile Client for iOS](#)

[Anyware Mobile Client for Android](#)

[Anyware Mobile Client for Chromebooks](#)

Zero clients:

[Anyware Tera2 Anyware Zero Client](#)

Additional Information

Information about **Anyware client connectivity requirements and usage instructions**, is available in the following documentation:

- Software clients:
 - [Anyware Software Client for Windows](#)
 - [Anyware Software Client for macOS](#)

- [Anyware Software Client for Linux](#)
- Mobile Clients:
 - [Anyware Mobile Client for iOS](#)
 - [Anyware Mobile Client for Android](#)
 - [Anyware Mobile Client for Chromebooks](#)
- Zero clients:
 - [Anyware Tera2 Anyware Zero Client](#)

Security Guide

Security in agents

Security Certificates in Anyware Agents

PCoIP requires a certificate to establish a session. By default, Anyware agents generate a self-signed certificate that secures the PCoIP session. Each component in the PCoIP system can generate these self-signed certificates, which will automatically work together without requiring any configuration.

You can, if needed, create and deploy your own custom certificates instead of relying on HP's self-signed certificates. This section explains how to create and implement custom certificates.

Using Custom Security Certificates

You can use OpenSSL, Microsoft Certification Authority, or a public certificate authority (CA) of your choice to create your certificates. If you are not using OpenSSL, consult your certificate authority's documentation for instructions on creating certificates in a Windows Certificate Store-compatible format.

The procedures in this section use OpenSSL to generate certificates that will satisfy most security scanner tools when the root signing certificate is known to them.

Caution: Certificates are stored in the Windows Certificate Store

Certificates are stored in the Windows certificate store. If you have old certificates that are stored on the host, they should be deleted to avoid conflicts or confusion.

Custom Certificate Guidelines

If you choose to use your own certificates, follow these general guidelines:

Save your root CA signing certificate in a safe place for deployment to clients.

Back up private and public keys to secure locations.

Never store files created when generating keys or certificates on network drives without password protection.

Once certificates have been deployed to the Windows certificate store, the files they came from are no longer needed and can be deleted.

Standard automatic tools, such as Automatic Certificate Enrollment and Group Policy, can be used for deploying automatically generated certificates. Both Automatic Certificate Enrollment and Group Policies are implemented through Active Directory. See MSDN Active Directory documentation for more information.

Pre-session Encryption Algorithms

Connections are negotiated using the following supported RSA cipher suites:

- TLS_ECDHE_ECDSA_WITH_AES_256_GCM_SHA384

- TLS_ECDHE_RSA_WITH_AES_256_GCM_SHA384
- TLS_AES_256_GCM_SHA384

 Note: Minimum SSL version

These Max Compatibility security level cipher suites have a minimum required SSL version of TLS 1.2.

Custom Security Certificates

In order for a CA signed certificate to be loadable by Anyware components, it must be stored in `/etc/pcoip-agent/ssl-certs` in three **.pem** files, owned by the `pcoip` user, and only readable by the owning user:

- **pcoip-key.pem** must contain an unlocked RSA key
- **pcoip-cert.pem** must contain a certificate that signs the key in `pcoip.pem`
- **pcoip-cacert.pem** must contain a CA certificate chain that validates the certificate in `pcoip-cert.pem`

Configure the Graphics Agent for Linux to use custom certificates

The Graphics Agent for Linux can be configured to look locally for certificates or to generate its own by setting the `pcoip.ssl_cert_type` directive in `pcoip-agent.conf`.

For more detailed information, see [Configuring the Agent](#).

Select a Security Key Length

When the Graphics Agent for Linux is attempting to find a certificate in storage, the required key length can be set via the `pcoip.ssl_cert_min_key_length` directive in `pcoip-agent.conf`.

If the system cannot find a local certificate with the specified key length, it will either self-generate a certificate (if `pcoip.ssl_cert_type` is 0), or refuse the connection (if `pcoip.ssl_cert_type` is 1). This setting has no effect if `pcoip.ssl_cert_type` is set to 2.

For more detailed information, see [Configuring the Agent](#).

Zero Trust via Trust Agent (Linux)

Trust Agent on Linux

Background

The Trust Agent is a background service that enables the Anyware Host Agent to connect to a Trust Center and enforce end-to-end Zero Trust policies.

- The service runs alongside the Host Agent.
- It becomes functional only after a local administrator connects it to a Trust Center using the setup flow described in [Enforcing Zero Trust Policies](#).
- Even if not configured, the service runs in the background and activates immediately once connected to a Trust Center.

Note: The Trust Agent is included with all Anyware Host Agent installation packages starting with version 25.10.

Data Storage Locations

Directory	Purpose
<code>/var/lib/anyware-trust-agent</code>	Persistent state (Trust Center registration, applied configuration, certificates, etc.)
<code>/etc/anyware-trust-agent/onboarding-token</code>	Onboarding token

- Logs are written to `syslog` and can be viewed using tools like `journalctl` .

To clear the persistent Trust Agent state after uninstall or to de-register it from your Trust Center, delete the directories listed above.

Known Issues

No known issues specific to Linux at this time.

Reference

Wacom Local Termination on Ubuntu Cloud Hosts

Cloud-based Ubuntu hosts will fail to properly identify Wacom tablets that have been locally-terminated at the Anyware client. When this occurs, pressure sensitivity and other advanced features will not work properly.

To work around this issue, remove the default AWS, Microsoft Azure, or Google Cloud kernel and replace it with a generic kernel.

Note: Ubuntu cloud hosts only

This procedure applies only to Ubuntu hosts on AWS, Microsoft Azure, or Google Cloud Platform. All valid RHEL and non-cloud Ubuntu installations work as expected.

To enable local termination:

1. First, confirm that you need to replace the kernel. Open a console and enter the following command:

```
uname -r
```

If the response contains the word `generic` (for example, `4.15.0-66-generic`) then your kernel is already generic and you can skip this procedure.

If the response ends in `aws` , `azure` , or `gcp` , note the version number and continue.

2. Find the available `linux-virtual` package for your distribution. In a console window, enter the following command:

```
apt-cache policy linux-virtual
```

In the response, note the candidate major version number. For example, if the candidate's number is `4.15.0.66.68` , then the major version number is `4` .

3. Compare the major versions of the *installed* kernel from step 1, and the *candidate* kernel in step 2:

- **If the major versions for the installed and candidate kernels are the same**

In a console window, enter the following command:

```
sudo apt install linux-virtual
sudo apt install linux-cloud-tools-virtual
```

- **If the major versions for the installed and candidate kernels are *not* the same**

1. Retrieve the full list of available kernels:

```
apt-cache policy linux-virtual*
```

Look through the output for the generic kernel version matching your installed kernel's major version.

2. Install the kernel and cloud tools packages for the correct version:

```
sudo apt install linux-virtual-<version>
sudo apt install linux-cloud-tools-virtual-<version>
```

...where `<version>` is the number reported in the output from `apt-cache policy linux-virtual*` .

For example, if you needed to find a kernel with a major version of `5` , you would look through the output of `apt-cache policy linux-virtual*` and find a response similar to this one:

```
linux-virtual-hwe-18.04:
  Installed: (none)
  Candidate: 5.0.0.32.89
  Version table:
     5.0.0.32.89 500
                500 http://ca.archive.ubuntu.com/ubuntu bionic-updates/main amd64 Packages
                500 http://security.ubuntu.com/ubuntu bionic-security/main amd64 Packages
```

The version is `hwe-18.04` . You will install that package and the corresponding cloud tools package:

```
sudo apt install linux-virtual-hwe-18.04
sudo apt install linux-cloud-tools-virtual-hwe-18.04
```

4. Purge the cloud-specific ubuntu image:

◦ **AWS:**

```
sudo apt purge linux*aws
```

◦ **Azure:**

```
sudo apt purge linux*azure
```

◦ **GCP**

```
sudo apt purge linux*gcp
```

5. When you see the *Abort kernel removal* message, respond with `No` .

6. Reboot the machine:

```
sudo reboot
```

7. When the machine comes back up, reconnect and check that the generic kernel is in use:

```
uname -r
```

You should see a response ending in `-generic` .

8. Obtain the `uhid` driver by installing `linux-modules-extra` for your kernel version:

```
sudo apt install linux-modules-extra-$(uname-r)
```

9. Reboot the machine:

```
sudo reboot
```

10. When the machine comes back up, reconnect and check that the uhid driver is present:

```
ls /dev/uhid
```

You should see a response similar to `/dev/uhid`.

11. Install USB driver packages:

```
sudo apt install usb-vhci-dkms
```

12. Reboot the machine:

```
sudo reboot
```

13. When the machine comes back up, reconnect and check that the USB drivers are present:

```
lsmod | grep usb
```

You should see a response similar to this:

```
usb_vhci_iocifc    20480 3
usb_vhci_hcd       20480 1 usb_vhci_iocifc
```

 Note: What if the response is empty?

If the output is empty, you may need to uninstall and reinstall the `vhci` package:

```
sudo apt remove usb-vhci-dkms
sudo reboot
sudo apt install usb-vhci-dkms
sudo reboot
```

14. Install the Wacom driver for your tablet.
15. Reboot the host machine.
16. If you have not installed the Graphics Agent for Linux, [install it now](#).

VSphere Displays

VSphere Reference

This section contains reference information useful for VSphere users configuring displays with the Anyware Graphics Agent.

You can find information here for:

- [Configuring multiple monitors for NVIDIA GRID vGPUs](#)
- [Configuring displays for NVIDIA GRID with pass-through](#)

Configuring Multiple Monitors for NVIDIA GRID vGPUs

Configuring Multiple Monitors for NVIDIA GRID vGPU

This section describes how to add a shared NVIDIA GRID PCI device to your virtual machine and specify its GPU profile. The following instructions are only required for virtualized workstations using a [supported NVIDIA GRID vGPU video card](#).

Note: Monitor limitations

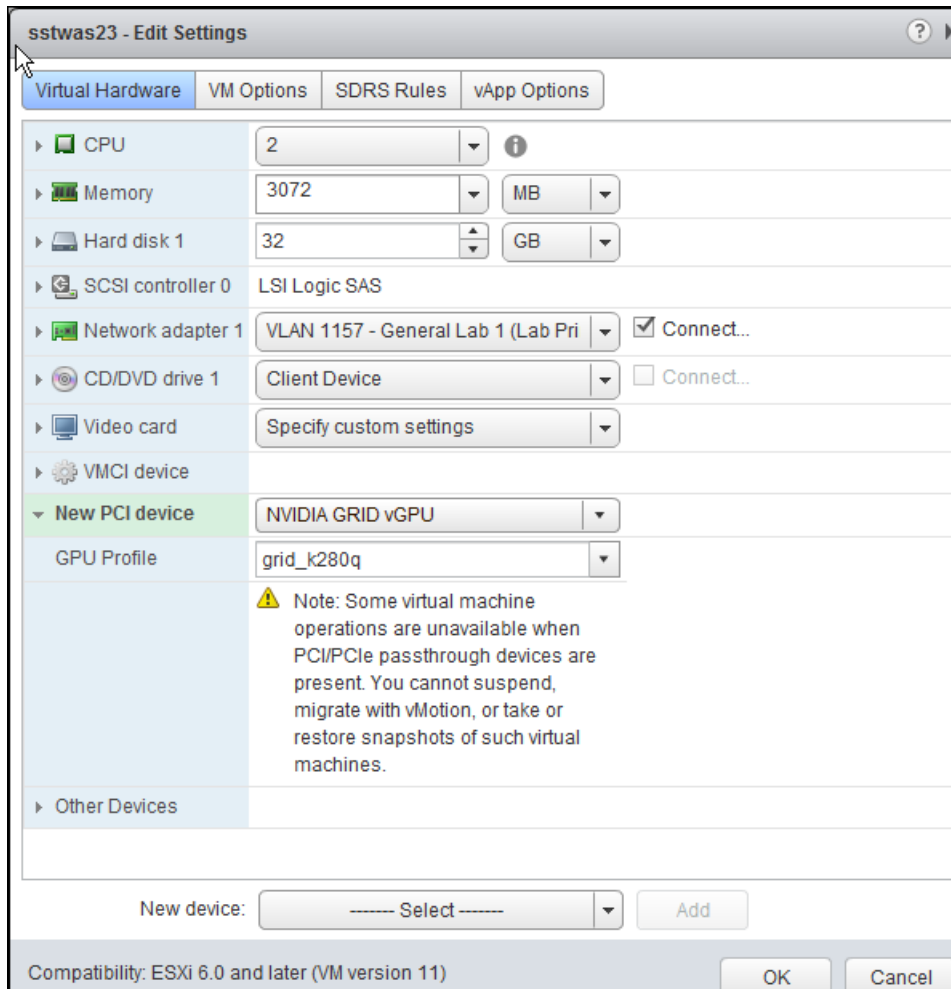
The Anyware Graphics Agent supports up to four displays. Your video card may limit the actual number of displays you can have.

Before you begin, ensure the following prerequisites are met:

- You have installed the NVIDIA GRID graphics card and supported driver on the ESXi host.
- You have installed VMware Tools on your ESXi host and the VMware SVGA 3D driver on the virtual machine.
- You have local administrative permission to the workstation.
- You have disabled OS power management features on the workstation.

To add a shared PCI device and specify its GPU profile:

1. Using vSphere Web Client, right-click the virtual machine in the Navigator list and select **Power > Power Off**.
2. Right-click the virtual machine and select **Edit Settings**.



3. In the New device drop-down list, select **Shared PCI Device** and then click **Add**.
4. Click **Reserve all memory**.
5. In the GPU Profile drop-down list, select the profile for your card. This profile determines how many virtual display heads, or displays, are available for your card.
6. Click **OK**.
7. Start the virtual machine.

Configuring Displays for NVIDIA GRID with Pass-Through

This section describes how to add an NVIDIA GRID PCI device to your ESXi host.

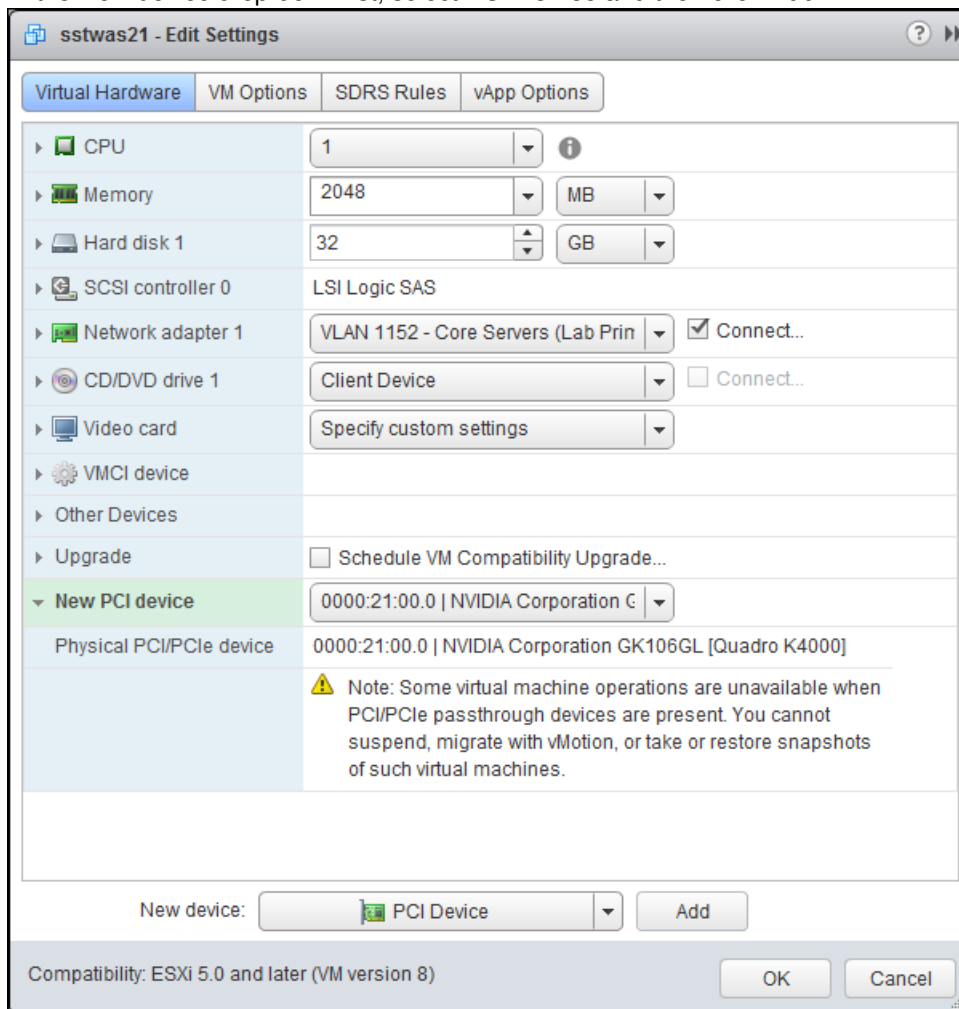
The following instructions are only required for virtualized workstations using a supported NVIDIA GRID video card with pass-through. For specific supported models, see [System Requirements](#).

Before you begin, ensure the following prerequisites are met:

- You have installed the NVIDIA GRID graphics card.
- You have installed VMware Tools on your ESXi host and the VMware SVGA 3D driver on the virtual machine.
- You have local administrative permission to the workstation.
- You have disabled OS power management features on the workstation.

To add a PCI device:

1. Using vSphere Web Client, right-click the virtual machine in the Navigator list and select **Power > Power Off**.
2. Right-click the virtual machine and select **Edit Settings**.
3. In the *New device* drop-down list, select **PCI Device** and then click **Add**.



4. Click **OK**.
5. Start the virtual machine.

Installing the Amazon Root CA 1-4 Certificate

Installing the Amazon Root CA 1-4 Certificates

Licensing issues with the Cloud License Service (CLS) might occur if the Amazon Root CA 1-4 certificates are not in the certificate bundle of your system. Therefore, updating your operating system is the quickest way to ensure that your system has these certificates.

⚠ Important

If updating your OS is not an option, install the root CA manually. For assistance with certificate updates, get in touch with [the Support team](#).

Installing the Amazon Root CA 1-4 certificates on RHEL/Rocky

1. Download the Amazon Root CA from the following location: <https://www.amazontrust.com/repository/>.
2. Copy the downloaded certificate file to the `anchors` directory found in the following location: `/etc/pki/ca-trust/source/`.
3. Update the trust list by running the following command:

```
sudo update-ca-trust
```

4. Run the following command to ensure that the `certs` directory has the **Amazon_Root_CA_*.pem (1-4)** files. The `certs` directory is available in the following location: `/etc/pki/ca-trust/source/`.

```
openssl x509 -in /etc/pki/ca-trust/extracted/pem/tls-ca-bundle.pem -text -noout
```

Installing the Amazon Root CA 1-4 certificates on Ubuntu

1. Download the Amazon Root CA from the following location: <https://www.amazontrust.com/repository/>.
2. Open a Terminal.
3. Run the following command to install the certificates:

```
sudo apt install ca-certificates
```

4. Update the trust list by running the following command:

```
sudo update-ca-certificates
```

5. Run the following command to ensure that the `certs` directory has the **Amazon_Root_CA_*.pem (1-4)** files. The `certs` directory is available in the following location: `/etc/ssl/certs`.

```
ls -lha /etc/ssl/certs | grep Amazon_Root_CA
```

Example Procedures for Setting up Linux Machines for Smartcard Authentication

Sample Procedures for Linux Smartcard Authentication Setup

This topic contains example procedures that illustrate how to configure Ubuntu and RHEL machines for smartcard authentication.

Ubuntu Machine Setup

Step I: Prepare the Machine

Info

This example assumes that the domain name is `example.net`.

1. Log on to the test machine.
2. Modify the hostname of the test machine to include the domain name:

```
sudo hostnamectl set-hostname <machine-name.example.net>
```

```
sudo reboot
```

3. Verify that the hostname has been updated successfully:

```
hostnamectl
```

4. Install the required packages:

```
sudo apt -y install realmd sssd sssd-tools libnss-sss libpam-sss adcli samba-common-bin oddjob oddjob-mkhomedir packagekit
```

5. Join the Active Directory domain, replace **SRV.WORLD** with **your domain name**, and replace **domain_join_user** with your **domain user** that has the necessary permissions to join the domain.

```
root@test:~# sudo realm join SRV.WORLD -U domain_join_user
```

Password for domain_join_user: # AD password

6. Verify that the **AD user information** is accessible:

```
root@test:~# id Serverworld@srv.world
uid=1259201103(serverworld@srv.world) gid=1259200513(domain users@srv.world) groups=1259200513(domain users@srv.world),1259200512(domain admins@srv.world),1259200572(denied rodc password replication group@srv.world)
```

```
root@test:~# nano /etc/pam.d/common-session
```

7. Add the following line at the end of **Step 5** to create a home directory on initial login:

```
session optional    pam_mkhomedir.so skel=/etc/skel umask=077
```

8. Sign in to the console or an SSH session with the domain user account to verify that the domain joining was successful.

Step II: Enable Smart Card Authentication

1. Install the **PKCS#11 Driver** for your smartcard:

1. Identify the PKCS#11 driver that supports your smart card device.
2. Enter the following command to install the OpenSC PKCS#11 driver:

```
sudo apt install opensc-pkcs11 -y
```

Note

If your device requires a different PKCS#11 driver, install the appropriate package instead.

2. Install **PCSCD** to enable the smart card reader.

```
sudo apt install pcscd -y
```

3. Extract the **CA chain** that signs the smart card user certificate, and include the **CA certificates** in the SSSD CA database:

```
sudo mkdir -p /etc/sssdpki -m 600
```

```
sudo su
cat test_ca_chain.pem >> /etc/sssdpki/sssdpki_auth_ca_db.pem
```

4. Set **pam_cert_auth** to "True" in the `/etc/sssdpki/sssdpki.conf` file.

```
[pam]
pam_cert_auth = True
```

5. Edit the `/etc/sssdpki/sssdpki.conf` file and add the following section. Replace `<your_domain_name>` with your actual domain name:

```
[certmap/<your_domain_name>/uon]
maprule=((userPrincipalName={subject_nt_principal})(samAccountName={subject_nt_principal.short_name}))
```

6. Restart the SSSD service:

```
systemctl restart sssd
```

7. Check if the SSSD service is running:

```
systemctl status sssd
```

Rocky Linux Linux 8/9 Machine Machine Setup

Step I: Prepare Your Machine

Info

This example assumes that the domain name is `example.net` .

1. Log on to the test machine.
2. Run the following command to modify the hostname of the test machine to include domain name:

```
sudo hostnamectl set-hostname <machine-name.example.net>
sudo reboot
```

3. Run the following command to verify that the hostname has been update successfully:

```
hostnamectl
```

4. Install the required packages for domain joining:

```
sudo yum install samba-common-tools realmd oddjob oddjob-mkhomedir sssd adcli krb5-workstation -y
realm discover example.net
sudo realm join example.net -U <domain admin that has permission to join domain>
```

5. Run the following command to verify that the domain joining was successful:

```
getent passwd <domain user name>
```

6. Verify that the domain user can access the machine:

```
ssh domain_user@example.net@machine_ip_address
```

Note

If the `ssh` session fails, check the system log for errors.

Step II: Enable Smart Card Authentication

1. Run the following commands to fix the `failed to verify CMS error` . For more information refer to [Red Hat Bugzilla - Bug 2060798](#).

This command only work on RHEL/Rocky Linux 9

```
'''
sudo update-crypto-policies --set DEFAULT:SHA1
sudo reboot
'''
```

This command works on both RHEL/Rocky Linux 8 and 9

```
""  
sudo dnf install opensc gnutls-utils krb5-pkinit -y  
""
```

2. Add the following to the `/etc/krb5.conf` file in the `[libdefaults]` section. For more information, refer to [MIT Kerberos Documentation](#).

```
pkinit_kdc_hostname = example.net
```

3. Add or modify the following configuration in the `/etc/sss/sssd.conf` file:

```
[domain/example.net]
```

Adjust this setting if the DoD CAC card does not work in the test environment:

```
krb5_auth_timeout = 30
```

4. Get the CA chain that signs the smartcard user certificate. Ensure that the CA chain is in the `pem` format.
5. Assume the smart card user cert is signed by **Ca-Auth-root-CERT.pem** and **Ca-Auth-leaf-CERT.pem**.
6. Include the CA certificates in the **SSSD CA database**:

```
sudo mkdir -p /etc/sss/pki -m 600 (Create /etc/sss/pki if it does not exist)  
sudo su  
cat Ca-Auth-root-CERT.pem Ca-Auth-leaf-CERT.pem >> /etc/sss/pki/sss_auth_ca_db.pem
```

7. Add the CA chain to the system's trust anchors:

```
sudo trust anchor /etc/sss/pki/sss_auth_ca_db.pem
```

8. Run the following command to restart SSSD to apply changes:

```
sudo systemctl restart sssd
```

9. Allow both the smartcard and password authentication with the following command:

```
sudo authselect select sssd with-smartcard --force
```

Troubleshooting and Support

Support

Contacting Support

If you encounter any problems installing, configuring, or running the Graphics Agent for Linux, you can create a [support ticket](#).

Before creating a ticket, be prepared with the following:

- A detailed description of the problem
- Your agent version number ([how do I find my version number?](#))
- A prepared [support file](#)
- The local time when the problem occurred, in the HH:MM:SS format

The Community Forum

The PCoIP Community Forum enables users to have conversations with other IT professionals to learn how they resolved issues, find answers to common questions, have peer group discussions on various topics, and access the PCoIP Technical Support Service team. The HP staff are heavily involved in the forums.

To visit the community, go to [HP Anyware Support Service](#).

Finding your Agent Version Number

Finding the Agent Version Number

To find the agent's version number in Ubuntu:

```
dpkg -l "pcoip*"
```

To find the agent's version number in RHEL or Rocky Linux:

```
rpm -qai "pcoip*"
```

The console will display a table of all registered components and their version number, if they have one.

Creating a Support File

Creating a Technical Support File

We may request a support file from your system in order to troubleshoot and diagnose issues. The support file is an archive containing Anyware Graphics Agent for Linux logs and other diagnostic data that can help support diagnose your problem.

To create a support file, type the following command as a super user:

```
sudo pcoip-support-bundler
```

The support file will be created and placed in your /tmp directory. A message will display containing the full system path to the generated file.

Troubleshooting

Performing Diagnostics

Each Anyware component creates and updates a log file which records its activity as the system is used. Most troubleshooting within a Anyware system begins by examining these log files and looking for error conditions or other indications that may explain why the system is not operating as expected.

Log files for the Graphics Agent for Linux and other Anyware components are saved to [specific directories](#).

Note: Bundling log files for support

When investigating issues with HP support, you may need to provide a support file which includes system log files. Instructions are provided [here](#).

Locating Agent Log Files

Log files for the Anyware agent are located in the following directories by default. If you changed your agent's location during installation, the log files will be in your custom location instead.

Component	Log file location
Agent	/var/log/pcoip-agent/agent.log
Session Launcher	/var/log/pcoip-agent/session-launcher.log
Server/User	/var/log/pcoip-agent/server.{ user }.log

Note: Bundling log files for support

When investigating issues with HP support, you may need to provide a support file which includes system log files. Instructions are provided [here](#).

Setting Log Verbosity

Each Anyware component generates diagnostic log messages. The default log levels are recommended for use in a production deployment. When troubleshooting a particular problem, Teradici Support Services may recommend adjusting the event log verbosity level to obtain more information from certain parts of the system.

Note: This is a global setting

The `pcoip.event_filter_mode` directive is a global setting, and affects the output levels of all Anyware components.

To change the log verbosity level, set the `pcoip.event_filter_mode` directive in the pcoip-agent.conf file. See [PCoIP event log](#)

[verbosity](#) for instructions.

Log rotation

Log files in Linux agents are managed by `logrotate` . To manage how log files are rotated, edit the following files:

- `/etc/logrotate.d/pcoip-*`
- `/usr/share/pcoip-agent/pcoip-server.logrotate`

Session Log IDs

At the start of each PCoIP session, a unique session ID is generated by the Anyware client and passed to all connected Anyware components (including the Graphics Agent for Linux). Log messages generated by the agent are prefixed with this session ID, making it easy to identify. All log messages generated during a single session, by any Anyware component, will be prefixed with the same session log ID in RFC-4122 format:

```
yyyy-mm-ddThh:mm:ss.fffffZ xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxx > ...
```

For example:

```
2015-11-06T08:01:18.688879Z 4208fb66-e22a-11d1-a7d7-00a0c982c00d > ...
```

Log messages that do not pertain to a specific session will show a string of zeroes in place of the session log ID number.

If a Anyware component does not receive a session log ID from the PCoIP client, or receives an invalid value, it will generate a new session log ID and distribute it to the other components in the system.

Troubleshooting License Issues

HP includes a license validation utility that scans your local system and any connected physical or cloud-based license servers for active licenses, and informs you of when your license subscription expires. For more information, see the [FAQS on Licensing HP Anyware](#) in the Knowledge Base.

To run the license validation tool, type:

```
pcoip-validate-license
```

For more detailed information on `pcoip-validate-license`, type:

```
man pcoip-validate-license
```

To list your licenses and their expiration status, type:

```
pcoip-list-licenses
```

For more detailed instructions on `pcoip-list-licenses`, type:

```
man pcoip-list-licenses
```

Tracking Usage Over Time

Local License Server users can use our open-source script, which displays the maximum HP Anyware license concurrent usage for a license server over time. For more information, refer to our [Github page](#).

Cloud Licensing users can write a short script that runs `pcoip-list-licenses` periodically (for example, every 60 minutes) on any Anyware agent machine to track license usage.

Frequently Asked Questions

Can I use a screensaver?

Yes. However, a blank, static screensaver will provide the most efficient CPU and network bandwidth usage.

Why can't I use the maximum number of monitors or monitor resolution?

GPU profile licensing, GPUs, or clients may limit monitor resolution and monitor count, which will prevent you from utilizing the full agent specification. For more information, see [supported displays](#).

Which graphics rendering APIs are supported?

The Graphics Agent captures and delivers the output from the GPU, which is responsible for providing rendering APIs like OpenGL. See the documentation for your GPU for more information about rendering APIs.

How quickly does an Anyware agent complete a connection?

Anyware agents can usually achieve a connection in 15 to 30 seconds. We use the statistical value Top Percentile (TP) to measure the time to establish a session:

- TP99: Ninety-nine percent of connections complete in under 30 seconds.
- TP50: Fifty percent of connections complete in under 15 seconds.

Why is my application not sending audio?

The Anyware agent delivers audio over PCoIP connections by reassigning the system's default audio device. Only applications that use the system default audio device will send or receive audio over PCoIP; applications that are configured to use non-default devices will not work. If you don't hear audio from your application, make sure it is configured to use the system default audio device.

I'm using Anyware Cloud Licensing. What network blocks should I leave open?

If you are using Anyware Cloud Licensing, you will need to add the following to your allowlist:

- teradici.flexnetoperations.com
- teradici.compliance.flexnetoperations.com

If you use an IP-based allowlist, we recommend your IT team add the following network blocks to your allowlist:

- IPv4: 185.146.155.64/27
- IPv6: 2620:122:f005::/56

⚠ Important: Migrating from the previous specification

Previously, our allowlist specification looked like this:

- **Production:** 64.14.29.0/24
- **Disaster Recovery:** 64.27.162.0/24

If you have an existing implementation using an IP-based allowlist like this, we recommend you leave it in place until the new allowlist is active and tested.

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